Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

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1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

The experiment investigates:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

In total, **17 models** were trained under one fixed protocol: the baseline and 16 attention variants, covering the five main experiments and three additional mechanisms (ECA, CBAM and Coordinate Attention).

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Dataset used: TensorFlow Flowers (`tf_flowers`)
The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images contain variations in scale, viewpoint, illumination, background clutter, flower size, color, texture and object location. These variations make the dataset suitable for studying the behavior of attention mechanisms.

The images were taken from the original archive that TensorFlow Datasets reads (`flower_photos.tgz`, 3,670 JPEG images): https://storage.googleapis.com/download.tensorflow.org/example_images/flower_photos.tgz. The class sizes are daisy 633, dandelion 898, roses 641, sunflowers 699 and tulips 799, so the classes are moderately imbalanced (largest/smallest = 1.42).

Hardware and software. All 17 models were trained in one notebook on an NVIDIA A40 GPU with TensorFlow 2.21.0, Keras 3.15.1 and Python 3.13. Each model took 14.0–14.5 minutes (241 minutes for all 17). The random seed (42) was set at the start and again before every model; GPU kernels were not forced to be deterministic, so repeating a run may change a few predictions (see Section 40). Every number in this report was read from the outputs of this single executed notebook.

3.1 Image Preprocessing

Every input image was resized (bilinear interpolation) to

$$224 \times 224 \times 3$$

and then passed through `keras.applications.resnet50.preprocess_input`, the preprocessing expected by the ImageNet-pretrained Keras ResNet50 (conversion from RGB to BGR and subtraction of the ImageNet channel means).

The following augmentation was applied to *training images only*:

- random horizontal flipping;
- random rotation (factor 0.05, i.e. about ±18°);
- random zoom (±10%);
- random translation (up to ±10% of the height and width); and
- random contrast variation (±10%).

Every model was given a freshly created augmentation pipeline with the same seed, so all models drew the same random stream. Validation and test images were only resized and preprocessed.

3.2 Dataset Split

The same training, validation and testing split was used for every experiment:

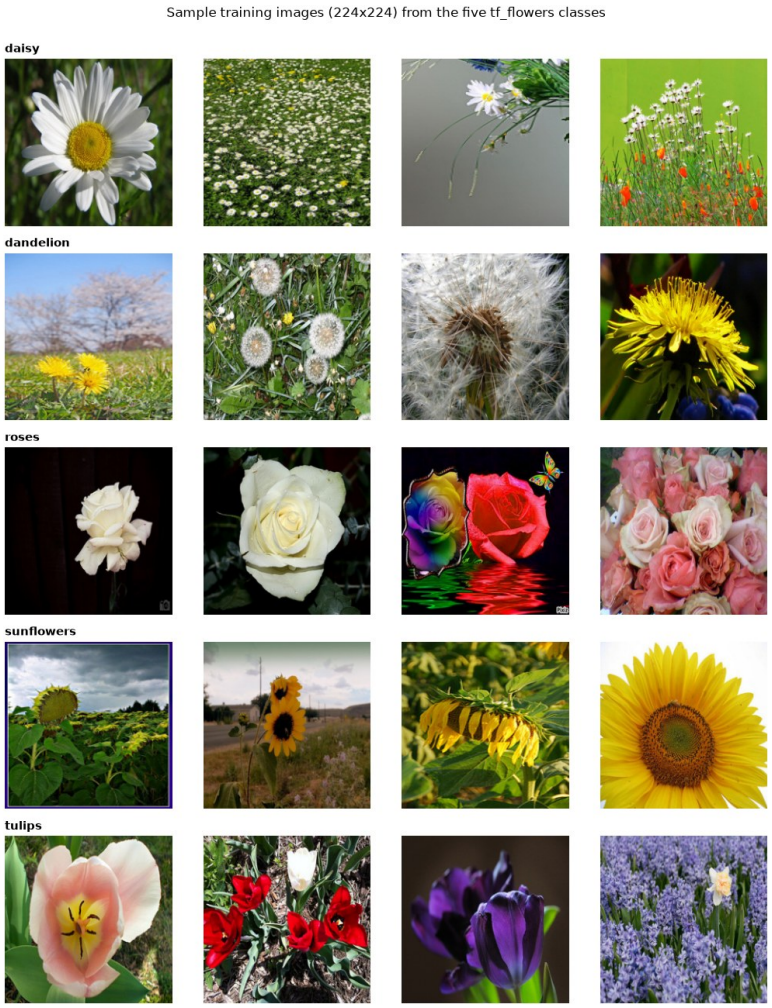
$$\boxed{70\% : 15\% : 15\%}$$

The split was stratified by class: the file list of each class was sorted, permuted with a seeded generator, and cut at 70% and 85%. It was created once in the notebook and reused by every model. The MD5 fingerprints of the three image arrays printed in the notebook (train `f7971466`, validation `2dddaabb`, test `dabcfc07`) identify it exactly.

Class	Train	Validation	Test	Total
daisy	443	95	95	633
dandelion	629	135	134	898
roses	449	96	96	641
sunflowers	489	105	105	699
tulips	559	120	120	799
Total	**2,569**	**551**	**550**	**3,670**

The validation set was used to choose, for each model, the epoch whose weights were kept (Section 20). The test set was used only for the final evaluation. Since the test set has 550 images, **one image corresponds to 0.18 percentage points of accuracy**; this scale is needed to judge the differences between models later in the report.

Plot 1: Sample Input Images from All Five Classes



Inference: the samples show why the dataset suits an attention study. Flower size ranges from a single close-up (rose, sunflower, tulip rows) to wide fields of tiny blossoms (daisy, dandelion rows); backgrounds include grass, sky, leaves and dark studio backdrops; and several classes resemble each other (red tulips and red roses, white daisies and dandelion seed heads). In many images, however, the flower fills most of the frame, so there is limited background that attention could suppress. This matters when the spatial-attention results are interpreted in Sections 28 and 29.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained on ImageNet (the Keras `ResNet50(include_top=False, weights="imagenet")` model) is used as the **only backbone architecture** in this experiment.

For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produced the following feature dimensions (Keras stores them as height × width × channels):

ResNet50 Stage	**Spatial Resolution**	**Channels**	**Output shape** (H, W, C)
Stem + MaxPool	56×56	64	(56, 56, 64)
conv2_x	56×56	256	(56, 56, 256)
conv3_x	28×28	512	(28, 28, 512)
conv4_x	14×14	1024	(14, 14, 1024)
conv5_x	7×7	2048	(7, 7, 2048)

The shapes of conv2_x to conv5_x are the shapes recorded by the notebook at the four insertion points (Section 13). As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow} \qquad \boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information. Each stage halves the grid, so one cell at conv5_x summarises a 32×32-pixel region of the input, whereas one cell at conv2_x summarises only 4×4 pixels.

The backbone without its ImageNet classifier has 23,587,712 parameters; the five-class head adds 10,245, so the baseline has 23,597,957 parameters.

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations. Section 29 (Plot 16) shows this directly for the spatial-attention maps learned at P_1–P_4.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.

Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of H, W, C and the semantic meaning of the feature tensor change significantly with network depth. Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

This experiment therefore asks both:

Which attention mechanism should be used?	Where should the attention mechanism be inserted?

7. Candidate Attention Positions

Four candidate attention positions were defined in ResNet50:

P_1 : after conv2_x, P_2 : after conv3_x, P_3 : after conv4_x, P_4 : after conv5_x.

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

The principal exercise concentrates on P_2, P_3 and P_4. Because the channel–spatial module is cheap, P_1 was also evaluated (model B1) as an extension. Self-attention was not placed at P_1 because its attention matrix would have 3136×3136 entries (Section 13).

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The original 1000-class ImageNet layer was removed. Global average pooling turns the $7 \times 7 \times 2048$ map into a 2048-vector, dropout ($p = 0.3$) is applied during training only, and a dense layer produces five class scores. The baseline reached **94.55% test accuracy (30 of 550 images wrong) and macro F1 94.59%**; every attention model is compared against these values.

9. Channel Attention

Consider a feature tensor $F \in \mathbb{R}^{H \times W \times C}$. Channel attention determines the relative importance of different feature channels. Global average pooling generates a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c), \qquad z \in \mathbb{R}^C.$$

The descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right), \qquad F_c = F \odot M_c(F),$$

where δ is ReLU, σ the sigmoid, W_1 reduces the channel representation and W_2 restores it. Channel attention answers:

$$\boxed{\text{Which feature channels are important?}}$$

Implementation. `ChannelAttention`: W_1 is a dense layer $C \to \max(C/16, 4)$ with ReLU and W_2 a dense layer back to C with sigmoid (reduction ratio $r = 16$, used throughout). At P_3 ($C = 1024$) this gives $1024 \cdot 64 + 64 + 64 \cdot 1024 + 1024 =$ **132,160** parameters. This module is the Squeeze-and-Excitation (SE) block, so model A1 serves as both "Channel Attention" and "SE".

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance. For $F \in \mathbb{R}^{H \times W \times C}$, average and maximum pooling are computed across channels and concatenated:

$$F_{\text{avg}} = \text{Mean}_c(F), \qquad F_{\max} = \text{Max}_c(F), \qquad F_s = [F_{\text{avg}} \| F_{\max}] \in \mathbb{R}^{H \times W \times 2}.$$

The spatial attention map and the refined feature are

$$M_s(F) = \sigma\left(\text{Conv}_{7\times7}(F_s)\right) \in \mathbb{R}^{H \times W \times 1}, \qquad F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

Implementation. `SpatialAttention`: one $k \times k$ convolution from 2 channels to 1 with "same" padding (so $H \times W$ is preserved) and a sigmoid. With $k = 7$ it has $2 \cdot 49 + 1 =$ **99** parameters, independent of the number of channels ($k = 3$: 19, $k = 5$: 51).

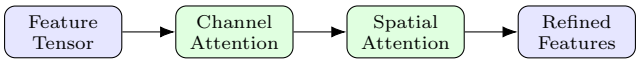
11. Sequential Channel–Spatial Attention

Channel and spatial attention were applied sequentially:

$$F_c = M_c(F) \odot F, \qquad F_{cs} = M_s(F_c) \odot F_c.$$

$$\boxed{F \to \text{Channel Attention} \to \text{Spatial Attention} \to F_{cs}}$$

This arrangement first determines "what is important" and then "where it is important."



Implementation. The two modules above applied one after the other (132,259 parameters at P_3). The reverse order (Spatial→Channel, Section 17) uses the same two modules in the opposite order and has the same number of parameters.

12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations. For $F \in \mathbb{R}^{H\times W\times C}$, the spatial dimensions are flattened to $X \in \mathbb{R}^{N\times C}$ with $N = HW$, and

$$Q = XW_Q, \qquad K = XW_K, \qquad V = XW_V,$$

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right), \qquad Y = AV, \qquad F_{\text{SA}} = F + \gamma\,\text{reshape}(Y),$$

where d is the query/key dimension and γ is a learnable scale. Self-attention answers:

Which spatial features are related to other spatial features?

Implementation. `SelfAttention` is the lightweight non-local form: W_Q and W_K are bias-free dense layers $C \to C/8$ ($d = 128$ at P_3), W_V maps $C \to C/2$, and, because V has only $C/2$ channels, a bias-free output projection W_O ($C/2 \to C$) restores the channel count, so that the residual is $F + \gamma$ reshape(YW_O). The scale γ is a single weight initialised to 0, so the module starts as an identity mapping. At P_3 this gives $2(1024{\cdot}128)+1024{\cdot}512+512{\cdot}1024+1 =$ **1,310,721** parameters, about ten times the channel module. Since A has size $N \times N$, its cost grows with the square of the number of locations, so self-attention was used at P_3 (Section 13).

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure is

$$\text{Previous ResNet Stage} \to \text{Attention} \to \text{Next ResNet Stage},$$

for example

conv4_x → Channel Attention → Spatial Attention → conv5_x

Implementation. The Keras ResNet50 was re-assembled layer by layer into a stem and four stages, re-using the original layer objects so that the ImageNet weights are untouched; after stage k the attention block registered for P_k (if any) is applied. One builder function therefore covers every experiment, and only a short plan such as {`3: cs`} or {`2: cs, 3: cs, 4: cs`} changes between models. Before any training, the notebook checked that the re-assembled baseline gives exactly the same feature map as the stock Keras ResNet50 (maximum absolute difference 0.0).

Initialisation of the attention gates. The gates use the default Keras initialisation: random weights, so a freshly created sigmoid gate outputs values around 0.5 and scales the pretrained features down at the start of training. Only the self-attention scale γ starts at exactly 0 (an identity mapping). This difference shows up in the early epochs of the training curves (Section 24.1 and 24.2).

Feature dimensions before and after each attention block. The notebook recorded the shape of the tensor entering and leaving every module at every position it was used, and a check before training built all nine module types at every position (34 combinations; self-attention and the combined module were skipped at P_1). In every case the input and output shapes were identical, so no module changes the spatial or channel dimensions:

Position	**Input** (H, W, C)	**Output** (H, W, C)	**Unchanged**
P_1	(56, 56, 256)	(56, 56, 256)	yes
P_2	(28, 28, 512)	(28, 28, 512)	yes
P_3	(14, 14, 1024)	(14, 14, 1024)	yes
P_4	(7, 7, 2048)	(7, 7, 2048)	yes

Parameters added by each module at each position (measured by building every module at every position; self-attention at P_1 was skipped on purpose):

Module	P_1: (56, 56, 256)	P_2: (28, 28, 512)	P_3: (14, 14, 1024)	P_4: (7, 7, 2048)
Channel / SE	8,464	33,312	132,160	526,464
Spatial (7×7)	99	99	99	99
Channel→Spatial and Spatial→Channel	8,563	33,411	132,259	526,563
Self-Attention	–	327,681	1,310,721	5,242,881
Combined (C→S→Self)	–	361,092	1,442,980	5,769,444
ECA	5	5	5	7
CBAM	8,563	33,411	132,259	526,563
Coordinate Attention	6,696	25,680	100,512	397,632

The channel-based modules grow with C^2 and become four times larger at each deeper stage; spatial attention is constant; self-attention grows fastest because of the $C \times C/2$ projections.

Size of the self-attention matrix at each position.

Position	$H \times W$	$N = HW$	Entries in A (N^2)	MB per image (fp32)	MB per batch of 32
P_1	56×56	3,136	9,834,496	37.5	1,200.5
P_2	28×28	784	614,656	2.34	75.0
P_3	14×14	196	38,416	0.147	4.69
P_4	7×7	49	2,401	0.009	0.29

Halving the resolution reduces the attention matrix 16 times. At P_1 the matrix alone would need about 1.2 GB per batch of 32 (before gradients), which is why self-attention was not evaluated there.

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention. The attention position was fixed at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.

Model	Channel	Spatial	Self-Attention
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

The following were kept identical for all six models: dataset split, preprocessing, augmentation, optimizer, learning-rate schedule, epochs, batch size, backbone, classifier, attention position and evaluation procedure (Sections 20 and 34). A5 applies Channel→Spatial→Self-Attention in that order.

Results (test set, 550 images).

Model	Acc. (%)	Prec. (%)	Recall (%)	Macro F1 (%)	Wrong (of 550)	Added params	ms/img	McNemar p vs A0
A0: Baseline	94.55	94.77	94.47	94.59	30	0	1.18	–
A1: Channel (SE)	**95.27**	**95.12**	**95.24**	**95.17**	**26**	132,160	1.21	0.454
A2: Spatial (7×7)	94.36	94.47	94.30	94.35	31	99	1.21	1.000
A3: Channel→Spatial	94.36	94.66	94.13	94.35	31	132,259	1.22	1.000
A4: Self-Attention	93.45	93.46	93.40	93.42	36	1,310,721	1.27	0.210
A5: C→S→Self	91.27	91.07	91.43	91.19	48	1,442,980	1.33	0.006

The last column is the exact McNemar test against the baseline A0 (Section 33.1); ms/img is the median batch-32 inference time.

Inference:

- **Best model.** Channel attention (A1, SE) is highest on every metric: 95.27% accuracy and 95.17% macro F1, against 94.55% and 94.59% for the baseline. This is +0.73 percentage points, i.e. **4 more correct images** (A1 corrects 10 baseline errors but breaks 6 images the baseline got right). With 550 test images this is not statistically distinguishable from the baseline (McNemar $p = 0.454$).
- **Spatial and channel+spatial attention do not help.** A2 and A3 both score 94.36%, one image below the baseline, with $p = 1.0$. Spatial attention alone adds only 99 parameters and does not change what the network learns in a useful way; its learned maps are examined in Section 28.2.
- **Self-attention lowers accuracy.** A4 reaches 93.45% (−1.09 pp, six more errors, $p = 0.21$) while adding 1.31 million parameters and 7% more inference time.
- **Combining everything is clearly worse.** A5 reaches only 91.27% (48 errors; −3.27 pp) and is the only model in the whole study whose difference from the baseline is statistically distinguishable ($p = 0.006$: the baseline is right and A5 wrong on 29 images, the reverse on 11). A5's best-validation checkpoint is epoch 5, the very end of the frozen phase: fine-tuning conv5_x never improved it on validation, and its training accuracy at that checkpoint is only 95.6% (all other models are at 98.1% or above), i.e. A5 is under-fitted rather than over-fitted. Stacking three untrained modules, two of which scale the pretrained features by a gate near 0.5 (Section 13), appears to leave too little signal for the head after only five frozen epochs.
- **Ranking.** A1 > A0 > A2 = A3 > A4 > A5 in accuracy, but apart from A5 every difference is at most 6 images and well inside the sampling noise of a 550-image test set (95% confidence intervals are about ±2 pp wide, Table in Section 33.1).

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention. The attention type was fixed as

Channel Attention → Spatial Attention

and exactly the same module was inserted at four different stages. B3 is the same trained model as A3 (it is not retrained), so the four positions are compared with one model per position.

Exp.	Attention position	Resolution	Channels	Acc. (%)	Macro F1 (%)	Wrong	Added params	ms/img	Train (min)	McNemar p
A0	none (baseline)	–	–	94.55	94.59	30	0	1.18	14.3	–
B1	After conv2_x	56 × 56	256	92.91	92.92	39	8,563	1.26	14.2	0.093
B2	After conv3_x	28 × 28	512	**94.73**	**94.70**	**29**	33,411	1.24	14.2	1.000
B3 (= A3)	After conv4_x	14 × 14	1024	94.36	94.35	31	132,259	1.22	14.0	1.000
B4	After conv5_x	7 × 7	2048	94.55	94.47	30	526,563	1.22	14.2	1.000

Inference:

- **Early attention is the worst.** Attention at P_1 (B1, 56 × 56) gives 92.91% accuracy and 92.92% macro F1, 9 images below the baseline ($p = 0.093$, the second smallest p-value in the study). It is also the slowest to train: its validation accuracy after epoch 1 is only 63.9% (baseline 87.5%) and it first exceeds 90% at epoch 7 (Section 24.2).
- **Middle and late positions match the baseline.** B2 (P_2) scores 94.73% (29 errors, the best of the four positions), B3 (P_3) 94.36% (31 errors) and B4 (P_4) 94.55% (30 errors, identical to the baseline). The 0.37 pp spread among P_2–P_4 corresponds to two images.
- **Trend.** Accuracy rises from P_1 to P_2 and then stays flat, while the module grows from 8,563 to 526,563 parameters (61 times). Low-level 56 × 56 features are the position where a freshly initialised gate does the most damage: a gate near 0.5 rescales 256 channels of edge and texture information that all later stages rely on, and with only five epochs of frozen training plus ten of fine-tuning conv5_x alone, the earlier unchanged stages cannot adapt to it (their weights stay frozen). This is an explanation consistent with the training curves, not something this experiment isolates.
- **Cost.** Inference time is 1.22–1.26 ms/img for all positions (+3 to +7% over the baseline); P_1 is the slowest of the four, probably because its feature maps have the most pixels, even though its module has the fewest parameters.
- **Answer to "how does the location of attention influence performance?"** It matters at the shallow end (P_1 hurts) and hardly at all from P_2 onward: on this dataset no position beats the baseline by a margin that can be told apart from noise.

16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage. The Channel→Spatial module was used at P_3 only, P_4 only, $P_3 + P_4$ and, as the extended experiment, $P_2 + P_3 + P_4$. Every position has its own module with its own weights.

Positions	Acc. (%)	Macro F1 (%)	Wrong	Added params	ms/img	Train (min)	Best epoch	McNemar p vs A0
Baseline (none)	**94.55**	**94.59**	**30**	0	1.18	14.3	14	–
P_3 only (A3)	94.36	94.35	31	132,259	1.22	14.0	12	1.000
P_4 only (B4)	**94.55**	94.47	**30**	526,563	1.22	14.2	13	1.000
P_3+P_4 (M34)	94.36	94.29	31	658,822	1.23	14.1	13	1.000
P_2+P_3+P_4 (M234)	93.64	93.64	35	692,233	1.25	14.3	12	0.383

Inference:

- **More locations did not help.** P_3+P_4 (94.36%, 31 errors) is exactly as accurate as P_3 alone and one image worse than P_4 alone, although it adds 659 thousand parameters (five times the P_3 module). Three positions (P_2+P_3+P_4) are worse still: 93.64% (35 errors) with 692 thousand parameters. M234 versus the best single position P_4: $p = 0.44$; versus the baseline: $p = 0.38$.
- **Convergence.** M234 needs 7 epochs to reach 90% validation accuracy (a single module at P_3 or P_4 needs 4 and 1) and has the largest final train–validation gap among the multi-position models (6.95 pp, Section 24.2). A likely reason is that repeated untrained gates compound the initial down-scaling of the features.
- **Cost.** Inference time grows from 1.18 ms (baseline) to 1.25 ms (three positions, +6%); training time per model stays at 14.1–14.3 minutes because the frozen backbone dominates.
- **Answer.** Attention at more locations is **not** necessarily more effective: the baseline and the single-module P_4 model tie for the best accuracy (94.55%), and the extra parameters, computation and memory of the multi-position models buy nothing measurable.

17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features. Both orders were trained at P_3 with identical modules (132,259 parameters each):

$$\boxed{\text{Channel} \to \text{Spatial}}\ (F \to M_c \to M_s) \qquad \text{versus} \qquad \boxed{\text{Spatial} \to \text{Channel}}\ (F \to M_s \to M_c).$$

Order	Acc. (%)	Prec. (%)	Recall (%)	Macro F1 (%)	Wrong	Params	ms/img
Baseline	94.55	94.77	94.47	94.59	30	0	1.18
Channel→Spatial (A3)	94.36	94.66	94.13	94.35	31	132,259	1.22
Spatial→Channel (O1)	**94.91**	**94.94**	**94.83**	**94.87**	**28**	132,259	1.22

Inference:

- Spatial→Channel (O1) is higher on all four metrics: 94.91% accuracy and 94.87% macro F1 against 94.36% and 94.35% for Channel→Spatial, a difference of 0.55 pp, i.e. **3 images** (28 versus 31 errors). It is also 0.36 pp above the baseline (2 images).
- Only 9 test images are classified differently by the two orders (3 where only C→S is right, 6 where only S→C is right); the exact test gives $p = 0.508$. The two orders are therefore **not** reliably different on this test set, and the sign of the difference could change with another seed.
- The learned spatial map does differ: when the spatial gate runs first it produces a much more contrasted map (mean raw value 0.937, contrast 0.852) than when it runs after the channel gate (0.985, 0.374; Section 28.2). A plausible reason is that the channel gate has already scaled the features, so the average and maximum pooled maps the spatial convolution receives have a smaller range. The classification result does not show that this matters.
- Both orders have the same parameters and the same inference time (1.22 ms/img).

18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result. The kernel size was varied over $k \in \{3, 5, 7\}$ in the Channel→Spatial module at P_3 (the 7×7 model is A3).

Spatial kernel	Accuracy (%)	Macro F1 (%)	Inference (ms/img)	Spatial-conv params	Module params	Wrong
3×3 (K3)	93.27	93.15	1.22	19	132,179	37
5×5 (K5)	94.18	94.11	1.22	51	132,211	32
7×7 (A3)	**94.36**	**94.35**	1.22	99	132,259	**31**

Inference:

- Accuracy increases monotonically with the kernel: 93.27% ($k = 3$), 94.18% ($k = 5$), 94.36% ($k = 7$), i.e. 37, 32 and 31 errors. Going from 3×3 to 7×7 gains 1.09 pp (6 images); going from 5×5 to 7×7 gains one image.

- The kernel size changes the module by only 80 parameters (19, 51 and 99) and the inference time by nothing measurable (1.22 ms/img for all three).
- $k = 3$ is also the slowest to converge (90% validation accuracy only at epoch 9) and under-fits the most (final training accuracy 98.1% versus 99.7–99.8% for $k = 5, 7$). Its spatial gate is the strongest: the mean raw gate value is only 0.742 and the max–min contrast 0.917, against 0.974/0.563 for $k = 5$ and 0.985/0.374 for $k = 7$ (Section 28.2). A 3×3 window sees only a very small neighbourhood of the 14×14 grid and, in this run, produced a gate that suppresses more activations.
- Larger kernels therefore give a small, useful amount of extra context, but the whole range from worst to best is 6 images and $k = 3$ versus $k = 7$ is not statistically separable ($p = 0.19$ for $k = 3$ against the baseline).

19. Additional Task: Comparison of Different Attention Mechanisms

To compare the laboratory mechanisms with established ones, **three** additional mechanisms were trained at the recommended position P_3 (after conv4_x) on the same ResNet50, same split and same protocol:

- **Efficient Channel Attention (ECA)**: global average pooling, then a 1-D convolution of kernel size 5 (no bias) across the channel descriptor, then a sigmoid; only 5 parameters.
- **CBAM**: a channel gate with a shared MLP applied to both the average- and max-pooled descriptors, followed by the 7×7 spatial gate (132,259 parameters at P_3).
- **Coordinate Attention**: pooling along height and width separately, a shared 1×1 convolution with reduction 32, batch normalisation and h-swish, then two 1×1 convolutions producing direction-aware gates (100,512 parameters; 100,448 trainable because of the batch-norm statistics).

The lightweight self-attention comparison is model A4 of Section 14.

19.1 Comparison

Attention	Acc. (%)	Prec. (%)	Recall (%)	Macro F1 (%)	Total params	Added	Inference (ms/img)
No Attention	94.55	94.77	94.47	94.59	23,597,957	0	1.18
Channel / SE	**95.27**	**95.12**	**95.24**	**95.17**	23,730,117	132,160	1.21
Spatial Attention	94.36	94.47	94.30	94.35	23,598,056	99	1.21
ECA	93.82	93.88	93.67	93.74	23,597,962	5	1.22
CBAM	93.45	93.59	93.40	93.44	23,730,216	132,259	1.23
Self-Attention	93.45	93.46	93.40	93.42	24,908,678	1,310,721	1.27
Coordinate Attention (additional)	93.27	93.23	93.02	93.10	23,698,469	100,512	1.23

"Added" is the number of parameters the module adds to the 23,597,957-parameter baseline.

19.2 Answers to the Attention-Mechanism Comparison Questions

1. **How does classification accuracy vary across attention mechanisms?** Between 93.27% (Coordinate Attention) and 95.27% (SE) at P_3 — a range of 2.0 pp or 11 images. Only SE is above the baseline (94.55%); spatial attention is 0.18 pp below, and ECA, CBAM, self-attention and Coordinate Attention are 0.7–1.3 pp below. None of the six differences from the baseline has $p < 0.2$ (Section 33.1).
2. **How does macro F1 vary?** It follows accuracy closely: 95.17% (SE), 94.59% (baseline), 94.35% (spatial), 93.74% (ECA), 93.44% (CBAM), 93.42% (self-attention), 93.10% (Coordinate). CBAM and self-attention tie in accuracy (93.45%) and macro F1 separates them by only 0.02 pp.
3. **How many parameters does each module add?** ECA 5; spatial 99; Coordinate 100,512; SE 132,160; CBAM 132,259; self-attention 1,310,721. The span is five orders of magnitude, and self-attention alone is 5.6% of the entire baseline network.
4. **How does each module influence inference time?** All add between 0.03 and 0.09 ms/img to the 1.18 ms/img baseline (+3 to +7%); self-attention is the slowest at 1.27 ms (+7.2%). ECA, with 5 parameters, is as slow as SE (1.22 versus 1.21 ms): the time is probably set by the extra operations and memory traffic on the $14 \times 14 \times 1024$ tensor rather than by the parameter count. Differences below about 0.05 ms are within the measurement noise noted in Section 22.
5. **Does attention improve the convergence behavior?** Not in general. Epochs needed to reach 90% validation accuracy: baseline 2, SE 3, spatial 2, CBAM 3, ECA 4, Coordinate 2, self-attention 1. Only self-attention is faster (it starts as an identity mapping, $\gamma = 0$, and reaches 92.0% validation accuracy after the first epoch); the modules with sigmoid gates start slower because they begin by scaling features by about 0.5. SE gave the smoothest fine-tuning of the main-experiment models (mean epoch-to-epoch validation change in phase 2: 0.67 pp, against 1.43 pp for the baseline). See Section 24.2.
6. **Does attention produce more focused feature localization?** Slightly, for some mechanisms. Measured on the Grad-CAM maps of the 550 test images (Section 27), the share of the image where the map exceeds half its

maximum is 24.5% for the baseline, 21.4% for CBAM, 21.8% for spatial and 22.7% for Channel→Spatial (more compact), but 28.8% for self-attention, 27.7% for the combined model and 32.1% for Coordinate Attention (more diffuse). Compactness does not predict accuracy.

7. **Does an increase in attention complexity always improve classification performance?** No. Rank correlation between the module's parameter count and test accuracy over the nine models A0–A5, ECA, CBAM and Coordinate Attention is $\rho = -0.55$ ($p = 0.13$): if anything, bigger modules did worse. The two largest modules (A4 with 1.31 M and A5 with 1.44 M parameters) are the two least accurate of the main experiment.
8. **Does a lightweight mechanism give comparable performance with a heavier one?** Yes. SE (132 K parameters) beats the 1.31 M-parameter self-attention by 1.82 pp (10 images) with one tenth of the parameters; ECA (5 parameters) is 0.37 pp *above* self-attention (93.82% versus 93.45%) with a quarter of a million times fewer parameters, and the spatial module (99 parameters) makes exactly the same number of errors (31) as the 132 K-parameter Channel→Spatial module.

The objective was to understand *performance + complexity + attention behaviour* rather than accuracy alone; on this problem the cheap mechanisms (SE, ECA, spatial) are the efficient choices and extra complexity is not rewarded.

20. Transfer-Learning Strategy

ImageNet-pretrained weights were used and the following strategy was applied to **every** model:

1. ResNet50 was loaded with ImageNet weights without its classification layer.
2. The selected attention module(s) were inserted at the chosen position(s).
3. Global Average Pooling, Dropout ($p = 0.3$) and a 5-neuron softmax Dense layer followed.
4. **Phase 1 (epochs 1–5):** the pretrained ResNet50 was frozen; only the attention modules and the classifier were trained with Adam, learning rate 10^{-3} constant.
5. **Phase 2 (epochs 6–15):** the final stage conv5_x was unfrozen (its batch-normalisation layers stayed frozen) and the network was fine-tuned with a new Adam optimizer and a cosine-decayed learning rate starting at 10^{-4} and ending at 0.

Batch size was 32 (81 steps per epoch), the number of epochs was fixed at 15 and **no early stopping** was used. After training, the weights of the epoch with the highest validation accuracy (ties broken by lower validation loss) were restored, over both phases; the test set was not used for this choice. The attention modules and the classifier are trainable in both phases; the backbone stages conv2_x–conv4_x stay frozen throughout. Trainable parameters are 10,245 for the baseline in phase 1 and 14.96 million in phase 2 (Section 33.2).

The hyper-parameters (learning rates, epochs, batch size, dropout, seed, optimizer) were identical for all 17 models, which keeps the comparisons controlled.

21. Loss Function

Categorical cross-entropy was used for all models. For N samples,

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where $y_{i,c}$ is the (one-hot) ground-truth label and $\widehat{y}_{i,c}$ the predicted probability. The same loss was used in every comparison. The test cross-entropy of the restored models ranges from 0.163 (SE) to 0.304 ($k = 3$); the baseline value is 0.209 (Section 38).

22. Evaluation Metrics

The following metrics were computed on the 550-image test set:

- classification accuracy;
- macro precision, macro recall and macro F1-score (scikit-learn);
- number of trainable parameters (phase 1 and phase 2) and total parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \qquad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c}, \qquad F1_c = 2\,\frac{\text{Precision}_c\,\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c},$$

and

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c, \qquad C = 5.$$

Inference time is the median over 5 passes over the whole test set at batch size 32 after a warm-up pass, divided by the 550 images. It is the throughput of the GPU, and the noise is a few percent (differences below about 0.05 ms/img should not be interpreted). The notebook also recorded single-image (batch size 1) latency, but it varies from 9.2 to 19.2 ms with no relation to the architecture (for example, the Spatial→Channel model and the Channel→Spatial model have identical operations and parameter counts but 9.17 versus 18.10 ms), so it is **not** used in any comparison.

Uncertainty. Each model was trained once. To avoid over-reading single numbers, the report adds (i) 95% bootstrap confidence intervals (1000 paired resamples of the test set, percentile method) and (ii) the exact McNemar test (a two-sided binomial test on the images that only one of two models classifies correctly) against the baseline (Section 33.1).

23. Required Main Result Table

Model	Position	Acc. (%)	Prec. (%)	Recall (%)	F1 (%)	Parameters (total)	Trainable (phase 2)	Inference (ms/img)
Baseline	–	94.55	94.77	94.47	94.59	23,597,957	14,963,717	1.18
Channel Attention	P_3	**95.27**	**95.12**	**95.24**	**95.17**	23,730,117	15,095,877	1.21
Spatial Attention	P_3	94.36	94.47	94.30	94.35	23,598,056	14,963,816	1.21
Channel+Spatial	P_2	94.73	95.04	94.57	94.70	23,631,368	14,997,128	1.24
Channel+Spatial	P_3	94.36	94.66	94.13	94.35	23,730,216	15,095,976	1.22
Channel+Spatial	P_4	94.55	94.55	94.43	94.47	24,124,520	15,490,280	1.22
Self-Attention	P_3	93.45	93.46	93.40	93.42	24,908,678	16,274,438	1.27
Additional Attention-1: ECA	P_3	93.82	93.88	93.67	93.74	23,597,962	14,963,722	1.22
Additional Attention-2: CBAM	P_3	93.45	93.59	93.40	93.44	23,730,216	15,095,976	1.23
Additional Attention-3: Coordinate	P_3	93.27	93.23	93.02	93.10	23,698,469	15,064,165	1.23

The position of the Channel→Spatial rows is P_2, P_3 and P_4 (B2, A3, B4); "Trainable (phase 2)" is the number of parameters updated during fine-tuning (conv5_x, the head and all attention modules). The best value in each column is in bold.

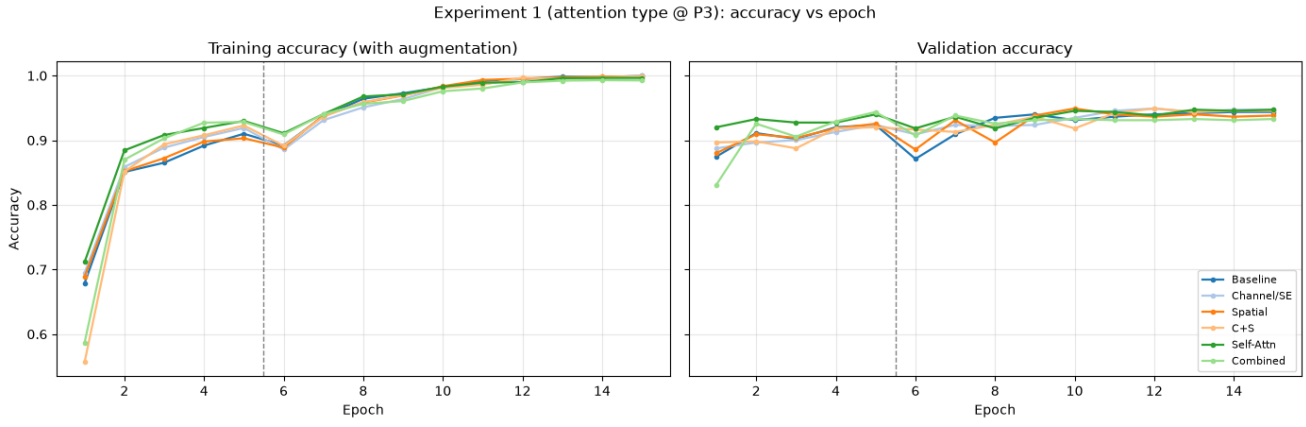
Inference: the table shows the central finding of the study: accuracy differences among all ten rows span only 2.0 pp (11 images) and are almost all within a few images of the baseline, while the added parameters span from 5 to 1.3 million. SE at P_3 is the best row on all four quality metrics, with a negligible 0.56% parameter increase, so it is the best accuracy–cost compromise in this run; but, given the 550-image test set, it should be described as *a small improvement that this experiment cannot confirm*, not as a proven one.

24. Graphical Performance Analysis

The numerical tables are complemented by the graphs below. All curves and bars were produced by the notebook from the same runs as the tables; "selected attention models" are the six models of the main attention-type experiment (Section 14) and, in Plot 4, also ECA, CBAM and Coordinate Attention.

24.1 Training and Validation Accuracy

Plot 2: Epoch versus Accuracy (Baseline, A1–A5)

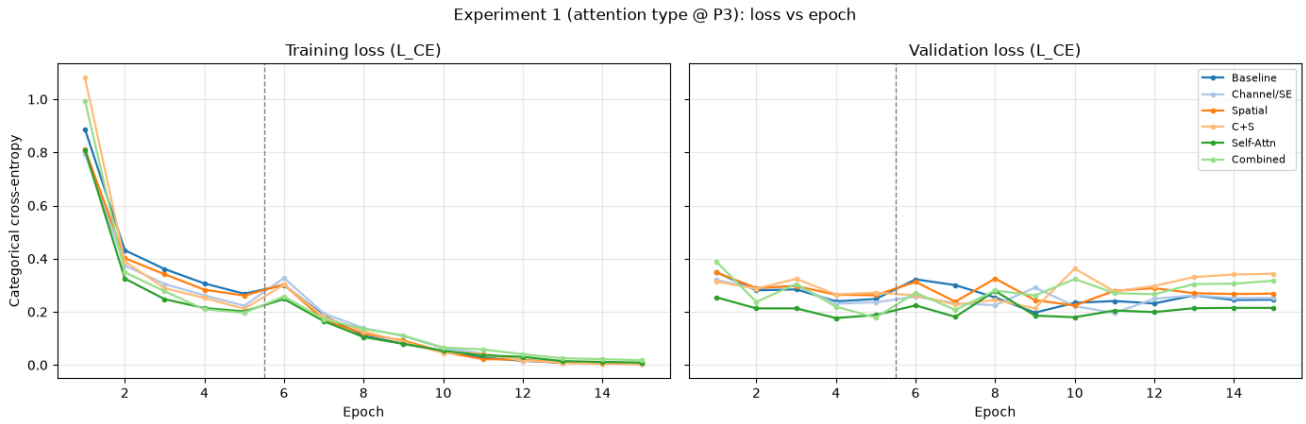


Inference: the dashed line marks the start of fine-tuning (epoch 6).

- **Convergence speed.** Self-attention (A4) starts best: 71.2% training and 92.0% validation accuracy after epoch 1 (baseline 67.9% and 87.5%), because its residual scale γ is initialised at 0 so it begins as the plain pretrained network. The sigmoid-gated variants start behind: Channel+Spatial (A3) trains at 55.7% and the combined model (A5) at 58.7% after epoch 1, and A5's validation accuracy is only 83.1%. All six models are within 4 pp of each other on validation by epoch 3 (88.8–92.7%), so attention did not accelerate convergence overall; it only removed the slow start in the self-attention case.
- **Smoothness.** The validation curves of every model zig-zag by 1–2 pp from epoch to epoch because the validation set has just 551 images (one image is 0.18 pp). The mean absolute epoch-to-epoch change during fine-tuning is lowest for SE (0.67 pp), then A3 (0.87), A5 (0.94), A4 (1.05), baseline (1.43) and spatial (1.94 pp): channel gating gave the smoothest fine-tuning.
- **The fine-tuning dip.** At epoch 6, when conv5_x is unfrozen, the baseline's validation accuracy falls by 5.3 pp (92.4% → 87.1%) and spatial attention's by 4.0 pp, whereas SE falls by 1.4 pp and Channel+Spatial by 0.4 pp. All models recover within two epochs.
- **Train–validation gap.** Training accuracy (measured on augmented images during training) reaches 99.3–99.96% at epoch 15 for all six models; the final-epoch gaps to validation accuracy are 4.9 (A4), 5.2 (A1), 5.4 (A3), 6.0 (A5) and 6.1 pp (A2) against 5.6 pp for the baseline, so no attention module reduced the gap in any meaningful way.
- **Overfitting and instability.** Every model fits the training set almost perfectly while validation accuracy plateaus at 93–95%, i.e. all are overfitted to the same degree. No model diverged, but several single-epoch validation dips occur (spatial attention at epoch 8: 89.7%; Channel+Spatial at epoch 10: 91.8%) – isolated, not a sign of training instability.

24.2 Training and Validation Loss

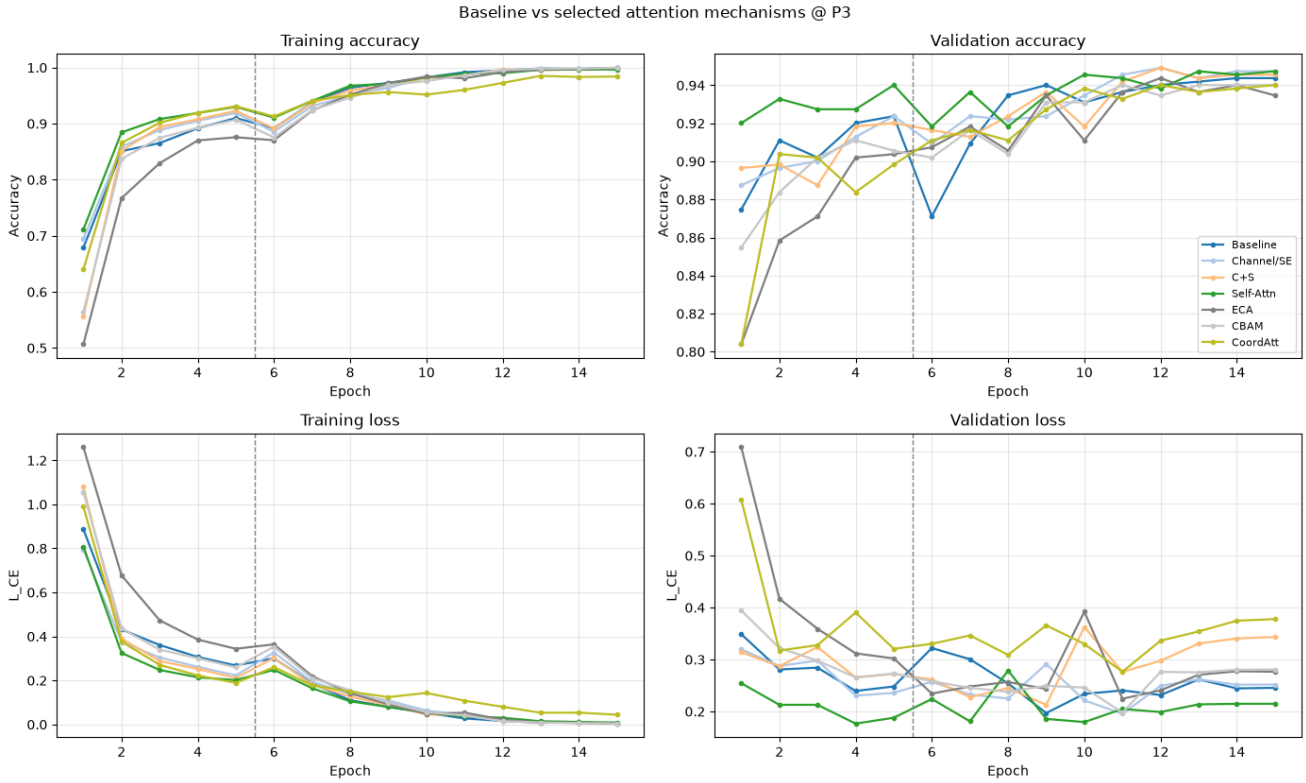
Plot 3: Epoch versus Cross-Entropy Loss (Baseline, A1–A5)



Inference:

- **Training loss** falls from 0.79–1.08 at epoch 1 to below 0.02 for every model at epoch 15. The highest starting values belong to Channel+Spatial (1.08) and the combined model (0.99) — the slow-start models — and the lowest to SE (0.79), self-attention and spatial attention (0.81), mirroring the accuracy curves.
- **Validation loss** is flat or slightly rising after epoch 5–10. Self-attention has the lowest validation loss for most epochs (0.18–0.28); the baseline reaches a minimum of 0.197 at epoch 9 and ends at 0.245. Channel+Spatial shows the clearest overfitting trend: from a minimum of 0.213 its validation loss rises to 0.343 (+0.13), with an isolated spike to 0.362 at epoch 10; the combined model shows the same rise (0.178 → 0.317 over the last epochs, although its restored checkpoint is epoch 5).
- **Loss versus accuracy.** The combined model has one of the *lowest* validation losses of all runs (0.178, at epoch 5, as low as self-attention's best of 0.177) yet the *worst* test accuracy (91.27%); lower cross-entropy does not guarantee higher accuracy when the model is under-fitted (its training loss at that checkpoint is still 0.124).
- **Conclusion.** The attention models differ in how fast they start, but all reach the same low training loss and the same validation-loss level (0.2–0.35), so the loss curves give no evidence that attention changes the final fit.

Plot 4: Convergence of the Baseline and the Additional Mechanisms (accuracy and loss, training and validation)



Inference:

- **Slow starters.** ECA is the slowest of the additional mechanisms: 50.7% training and 80.4% validation accuracy after epoch 1 (validation loss 0.71), and 87.6% training accuracy at epoch 5. Coordinate Attention also starts at 80.4% validation accuracy (validation loss 0.61), and CBAM at 85.5%. These modules start with randomly initialised sigmoid or h-swish gates, so the pretrained features are scaled down before the head has learned anything. Self-attention (92.0%) does not have this problem. By epoch 5 all nine models are between 89.8% and 94.4% on validation.
- **Epochs to 90% validation accuracy:** self-attention 1, baseline 2, Coordinate 2, CBAM 3, ECA 4. The validation accuracy at the end of the frozen phase is 92.4% (baseline), 90.6% (CBAM), 90.4% (ECA) and 89.8% (Coordinate): the additional mechanisms are *behind* the baseline after phase 1.
- **Late behaviour.** Coordinate Attention fits the training set least well (98.4% training accuracy and training loss 0.044 at epoch 15, against ≥99.3% for the others) and its validation loss rises from 0.277 to 0.378 at the end. ECA has an isolated validation-loss spike at epoch 10 (0.392). CBAM has the lowest validation loss of the three late in training (0.198 minimum at epoch 11), but ends on 93.45% test accuracy.
- **Fine-tuning smoothness.** The mean epoch-to-epoch change of validation accuracy in phase 2 is 1.29 pp for ECA, 0.78 for CBAM and 0.71 for Coordinate attention, against 1.43 for the baseline.

Convergence statistics for all 17 models. The table summarises the training curves of every model: the first epoch at which the validation accuracy is at least 85% and 90%; the validation accuracy at the end of the frozen phase (epoch 5); the roughness of fine-tuning (mean absolute epoch-to-epoch change of validation accuracy in epochs 6–15); the minimum and final validation loss, and the rise from the minimum to the final value; and the final train–validation accuracy gap.

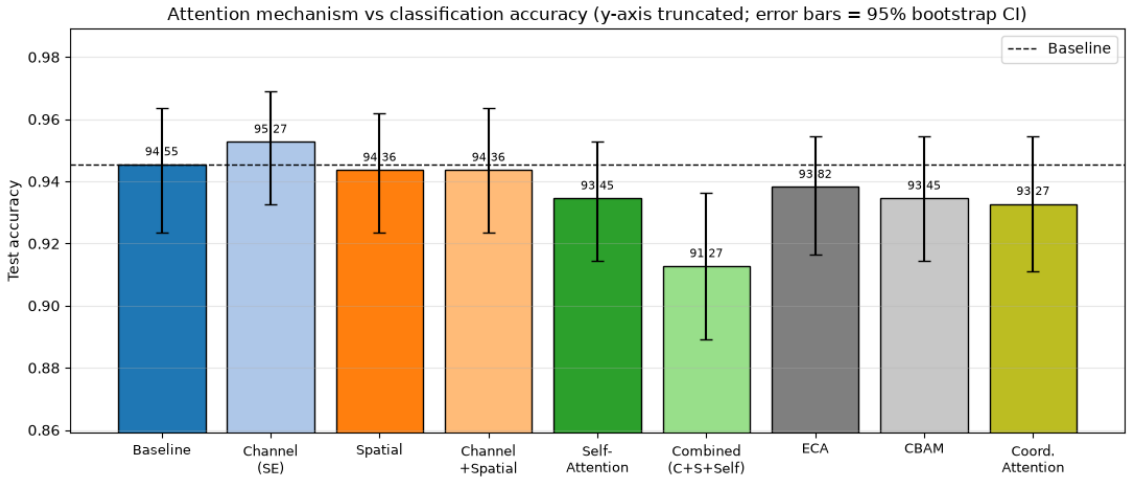
Model	Epoch val≥85%	Epoch val≥90%	Val acc at end of phase 1 (%)	Best epoch	Phase-2 roughness (pp)	Min val loss	Final val loss	Loss rise from min	Final train–val gap (pp)
A0: Baseline	1	2	92.38	14	1.43	0.197	0.245	0.049	5.59
A1: Channel (SE)	1	3	92.38	12	0.67	0.196	0.252	0.056	5.22
A2: Spatial (7×7)	1	2	92.56	10	1.94	0.224	0.268	0.044	6.05
A3: Channel→Spatial	1	4	92.01	12	0.87	0.213	0.343	0.130	5.37
A4: Self-Attention	1	1	94.01	13	1.05	0.177	0.215	0.038	4.91
A5: C→S→Self	2	2	94.37	5	0.94	0.178	0.317	0.139	5.98
B1: C→S at P_1	3	7	87.66	11	0.91	0.283	0.334	0.051	7.43
B2: C→S at P_2	2	4	90.02	12	0.65	0.230	0.313	0.084	6.77
B4: C→S at P_4	1	1	91.83	13	0.71	0.264	0.294	0.030	6.13
M34: C→S at P_3+P_4	2	3	92.38	13	1.13	0.232	0.277	0.045	5.81
M234: C→S at P_2+P_3+P_4	3	7	89.66	12	0.49	0.272	0.370	0.098	6.95
O1: Spatial→Channel	1	1	90.93	12	0.80	0.243	0.330	0.087	6.01
K3: C→S, $k = 3$	3	9	89.11	13	1.02	0.290	0.412	0.122	6.03
K5: C→S, $k = 5$	2	4	90.74	11	0.80	0.224	0.438	0.214	6.36
X1: ECA	2	4	90.38	12	1.29	0.225	0.277	0.052	6.46
X2: CBAM	1	3	90.56	11	0.78	0.198	0.281	0.083	5.95
X3: Coordinate Attention	2	2	89.84	12	0.71	0.277	0.378	0.101	4.39

Inference: the models that put sigmoid gates in the shallow, high-resolution stages are the slowest to converge: P_1 (B1) and the three-position model (M234) need 7 epochs to reach 90% validation accuracy and the 3×3-kernel model needs 9, against 1–4 for most single-module P_3/P_4 models and 2 for the baseline. These are also the models

with the largest train–validation gaps (B1 7.4, M234 7.0 pp, against 5.6 pp for the baseline). The largest rises of the validation loss from its minimum (K5 +0.21, A5 +0.14, A3 +0.13) are the clearest signs of overfitting; only four models have a smaller final gap than the baseline: SE (5.2), self-attention (4.9), A3 (5.4) and Coordinate Attention (4.4 pp), and the differences are ≤ 1.2 pp.

24.3 Attention Mechanism versus Accuracy

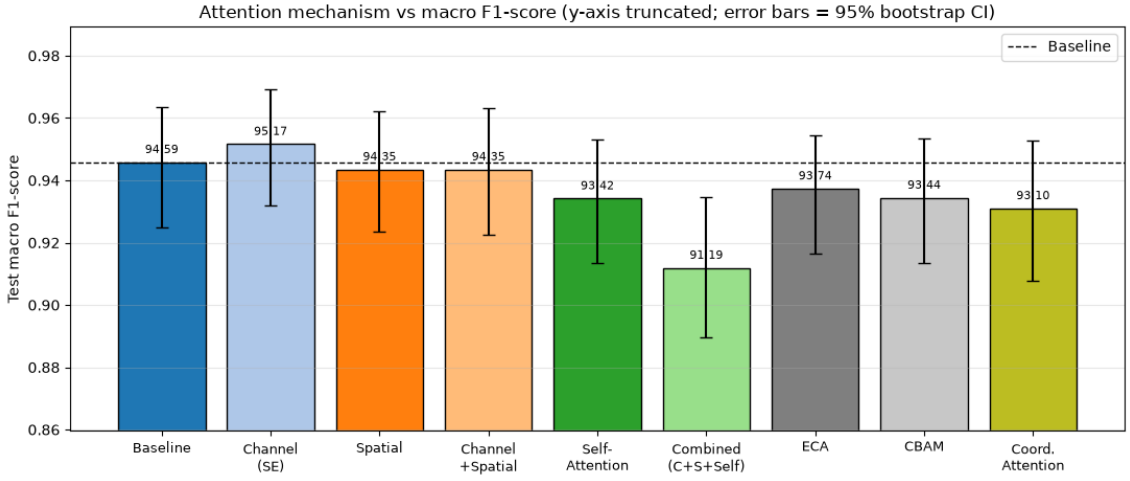
Plot 5: Attention Mechanism versus Test Accuracy



Inference: accuracy is 95.27% for SE, 94.55% for the baseline, 94.36% for spatial and Channel+Spatial, 93.82% for ECA, 93.45% for CBAM and self-attention, 93.27% for Coordinate Attention and 91.27% for the combined model. Only SE is above the baseline; the entire group from spatial to Coordinate Attention is within 1.3 pp (7 images) of the baseline, while the combined model stands apart at −3.27 pp. The vertical spread among the first eight bars (2.0 pp) is about the width of a single 95% confidence interval (about ±2 pp), so the ranking should not be over-read.

24.4 Attention Mechanism versus Macro F1

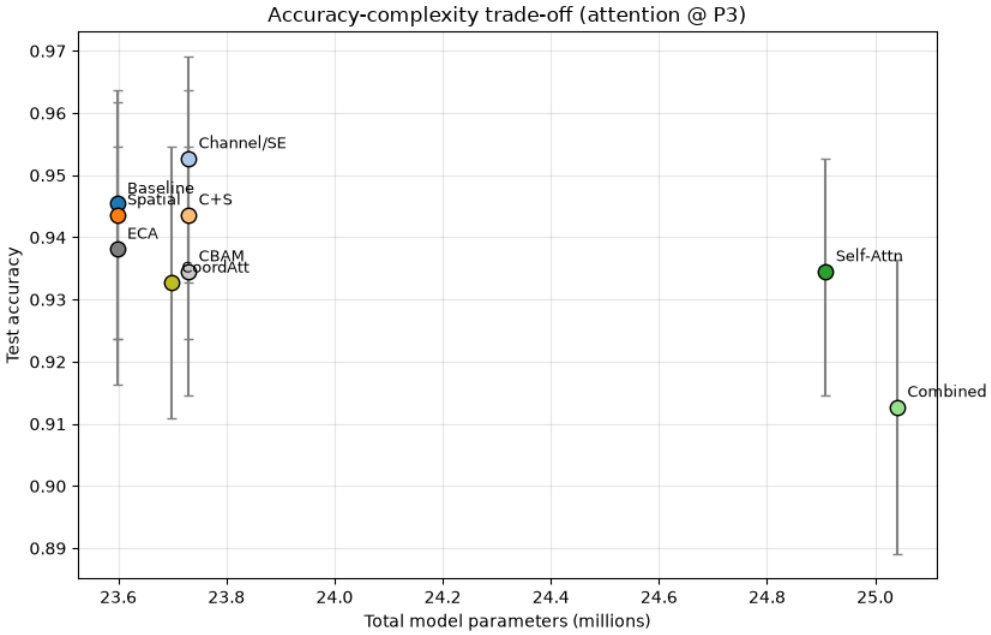
Plot 6: Attention Mechanism versus Macro F1



Inference: macro F1 reproduces the accuracy ordering (95.17% SE, 94.59% baseline, 94.35% spatial and C+S, 93.74% ECA, 93.44% CBAM, 93.42% self-attention, 93.10% Coordinate, 91.19% combined) and is never more than 0.2 pp from accuracy. The class counts are only moderately unbalanced (daisy 95 to dandelion 134 test images), so macro F1 adds little information here; it would diverge from accuracy if a mechanism favoured the large classes at the expense of the small ones, which none did. Per-class behaviour is analysed in Section 25.

24.5 Accuracy versus Model Parameters

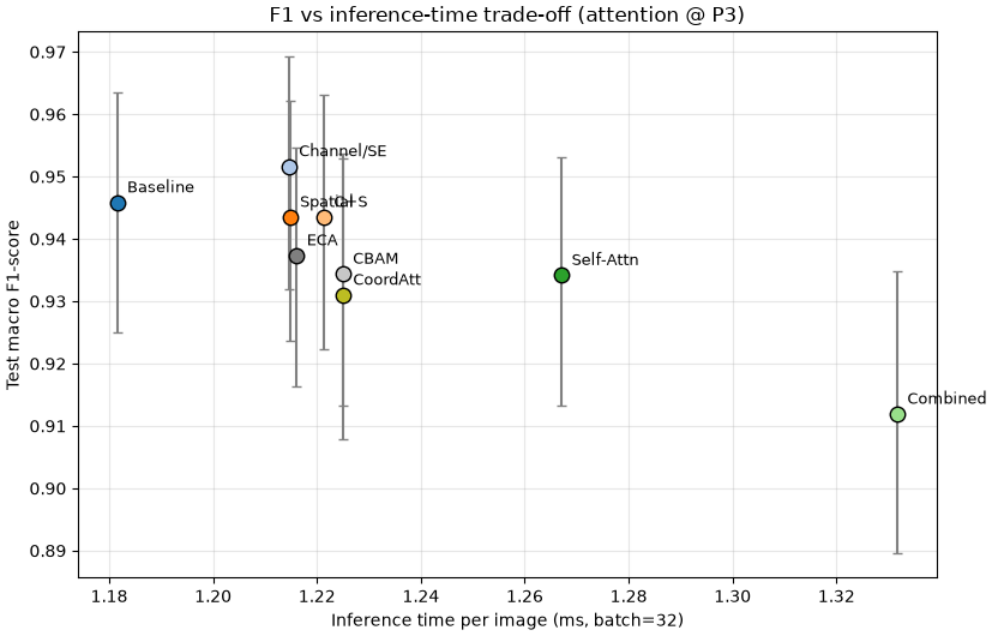
Plot 7: Test Accuracy versus Total Parameters (error bars: 95% bootstrap CI)



Inference: seven of the nine models sit in a vertical column between 23.60 M and 23.73 M parameters (baseline, spatial, ECA, SE, C+S, CBAM, Coordinate), because their modules add 0.6% or less to ResNet50, and accuracy within that column varies by 2.0 pp. The two models with self-attention move 1.3–1.4 M parameters to the right and *lower* (93.45% and 91.27%). There is thus no positive accuracy–complexity relationship (Spearman $\rho = -0.55$, $p = 0.13$ for the nine models), and the best trade-off is on the left of the plot: SE gives the highest accuracy for +0.56% parameters; ECA and spatial attention cost essentially nothing but also gain nothing. The error bars (about ±2 pp) of all nine models overlap, and the interval of the combined model (88.9–93.6%) only just reaches the lower end of the baseline's (92.4–96.4%); this agrees with the McNemar tests.

24.6 Macro F1 versus Inference Time

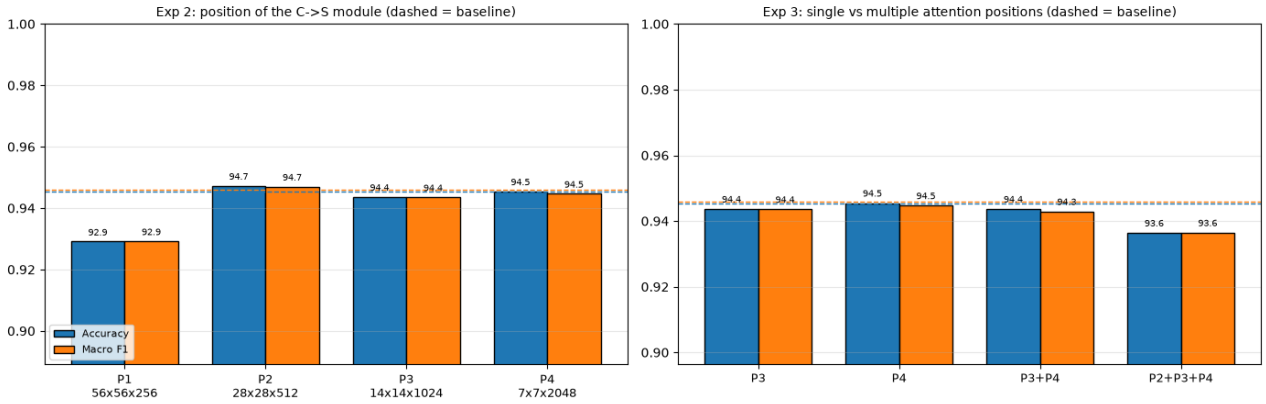
Plot 8: Macro F1 versus Inference Time per Image (batch size 32)



Inference: the baseline is fastest (1.18 ms/img). Six models cluster between 1.21 and 1.23 ms (+3 to +4%), self-attention is at 1.27 ms (+7%) and the combined model at 1.33 ms (+13%). Higher cost does not buy a higher F1: along the time axis F1 falls from 95.17% (SE, 1.21 ms) to 93.42% (self-attention, 1.27 ms) and 91.19% (combined, 1.33 ms). The extra computation of the heavier mechanisms is therefore not justified on this task. The differences within the main cluster are at the noise level of the timing measurement (a few percent).

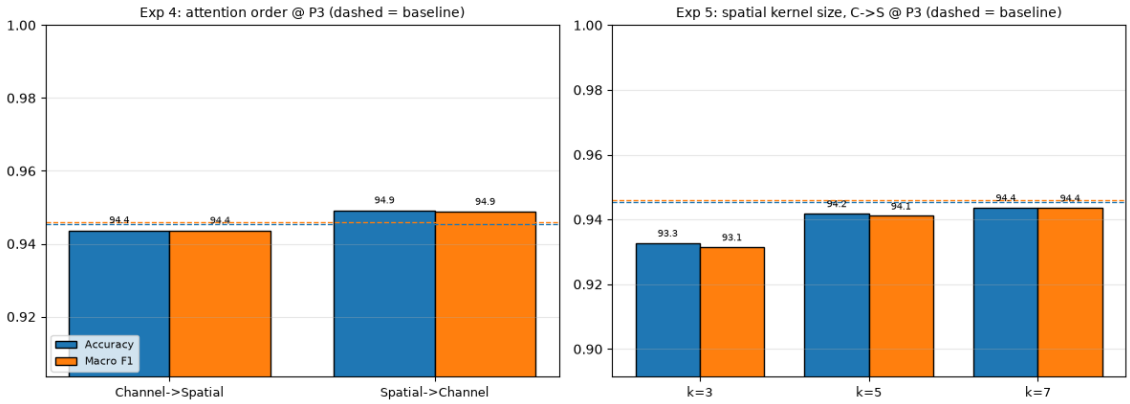
24.7 Comparison of Attention Position, Multiple Positions, Order and Kernel Size

Plot 9: Attention Position (left) and Single versus Multiple Positions (right)



Inference: the dashed lines are the baseline (94.55% accuracy and 94.59% macro F1). On the left, accuracy rises from 92.9% (P_1) to 94.7% (P_2) and then stays at 94.4–94.5% for P_3 and P_4, so only the earliest position falls clearly below the baseline. On the right, one position (P_3 94.4%, P_4 94.5%) is as good as two (P_3+P_4 94.4%) and better than three (93.6%): more attention locations lower rather than raise accuracy.

Plot 10: Attention Order (left) and Spatial Kernel Size (right)

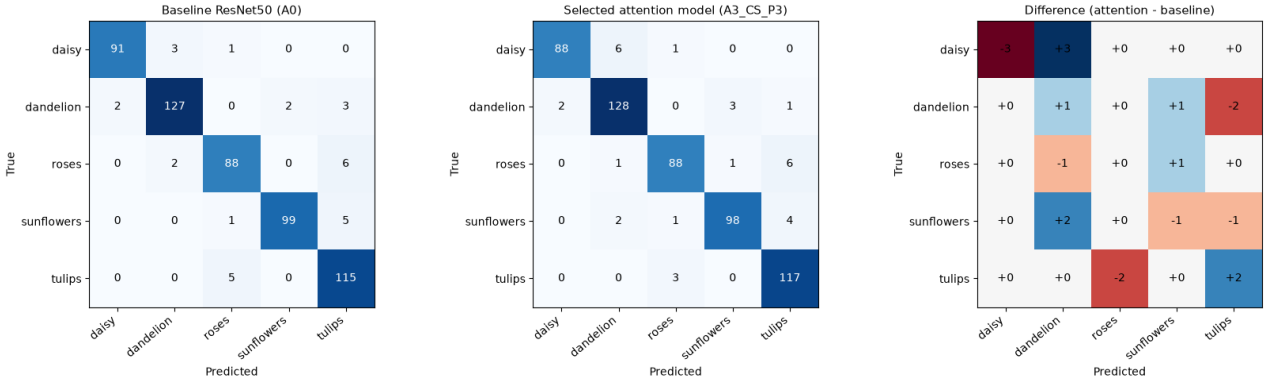


Inference: Spatial→Channel (94.9%) is above Channel→Spatial (94.4%) and the baseline, but the 3-image gap is not significant ($p = 0.51$). For the kernel size, accuracy rises monotonically with k (93.3%, 94.2%, 94.4%) and macro F1 follows (93.1%, 94.1%, 94.4%); only $k = 3$ is clearly below the baseline line.

25. Confusion Matrix Analysis

The confusion matrices of the baseline (A0) and the **selected attention model**, Channel+Spatial at P_3 (A3), are shown. The selected model was fixed *before* training as the representative attention model of the experiment; it is not the best-performing one (that is SE, A1), so Plot 11 shows a typical rather than a favourable case.

Plot 11: Confusion Matrices of the Baseline and the Selected Attention Model (test set)



Per-class results (test images: daisy 95, dandelion 134, roses 96, sunflowers 105, tulips 120):

Class	Recall A0 (%)	Recall A3 (%)	Precision A3 (%)	F1 A0 (%)	F1 A3 (%)	Change in F1 (pp)
daisy	**95.79**	92.63	97.78	**96.81**	95.14	−1.67
dandelion	94.78	**95.52**	93.43	**95.49**	94.46	−1.03
roses	91.67	91.67	94.62	92.15	**93.12**	+0.97
sunflowers	**94.29**	93.33	96.08	**96.12**	94.69	−1.43
tulips	95.83	**97.50**	91.41	92.37	**94.35**	+1.98

Inference:

- **Most accurately classified classes.** For the baseline, daisy (95.8% recall, F1 96.8%) and tulips (95.8%) are best and sunflowers has the best F1 after daisy (96.1%); roses is the weakest (91.7% recall, F1 92.2%).
- **Frequent confusions.** The dominant error is between the two red/pink flowers: roses → tulips (6 of 96 roses) and tulips → roses (5 of 120) for the baseline; sunflowers → tulips (5) is the next one, and daisy → dandelion (3). Tulips is an "attractor" class: 14 images of other classes are wrongly labelled tulips by the baseline (precision 89.1%).
- **Which class benefits from attention?** Tulips: recall rises from 95.8% to 97.5% (115 → 117 correct) and F1 from 92.4% to 94.4% (the largest gain, +2.0 pp), because tulip false positives fall from 14 to 11 (roses → tulips stays at 6 but tulips → roses falls from 5 to 3). Roses F1 also improves slightly (+1.0 pp).
- **Does attention reduce a particular type of confusion?** Yes, mainly tulips → roses (5 → 3) and sunflowers → tulips (5 → 4); but it *creates* errors elsewhere: daisy → dandelion doubles from 3 to 6 and sunflowers → dandelion goes from 0 to 2.
- **Is the accuracy change distributed across classes?** The model-level result (31 versus 30 errors) hides a redistribution: the F1 of two classes rises (tulips, roses) and that of three falls (daisy −1.7, sunflowers −1.4, dandelion −1.0 pp). The same total number of errors is therefore made in different places, which is why overall accuracy did not change.

26. Qualitative Analysis of Attention

Classification accuracy says nothing about where the network finds its evidence. The qualitative analysis therefore contains: Grad-CAM explanations of the baseline and of nine attention models (Section 27 and 30–32); internal attention maps (channel gates, spatial gates, self-attention; Section 28 and 29); and feature maps before and after attention (Section 28.1). The **same five test images** (indices 258, 460, 59 correct and 19, 334 wrongly classified by the baseline) are used in every comparison, as the manual asks.

27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction. Let A^k represent feature map k from a selected convolutional layer. For class score y^c, the importance coefficient is

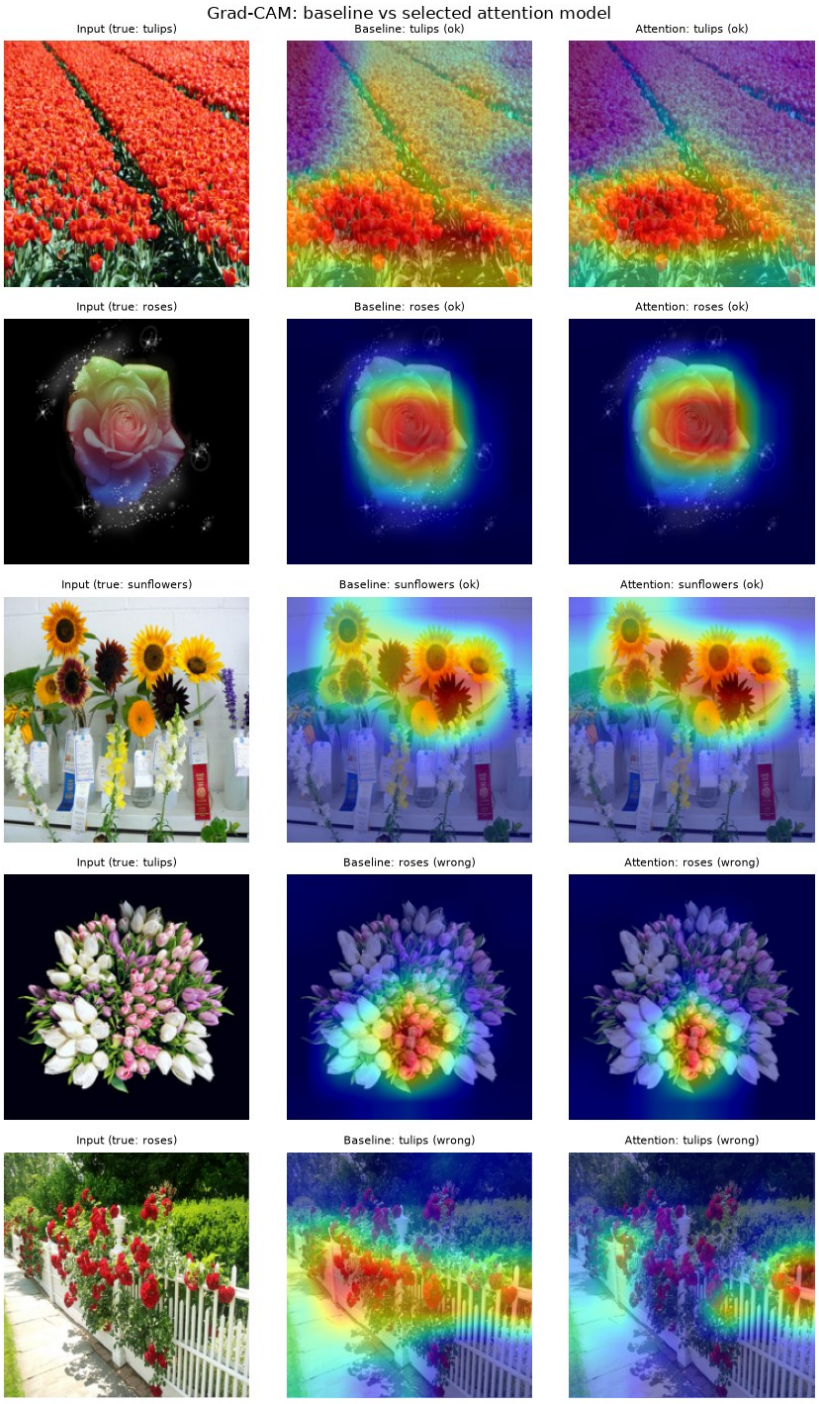
$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is

$$L^c_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

Implementation. The gradient of the predicted-class logit was taken with respect to the final $7 \times 7 \times 2048$ feature map (the output of conv5_x, i.e. after attention for P_4 models). The map is resized to 224×224 and overlaid on the input. The Grad-CAM is for the class the model *predicted*, which is why the label of a wrong prediction in the figures is the wrong class. Because the map has only 7×7 cells, each cell covers 32×32 pixels and the maps are inevitably coarse.

Plot 17: Input | Baseline Grad-CAM | Attention-Model (A3) Grad-CAM



27.1 Grad-CAM Interpretation

Inference (what is visible in Plot 17):

- **Correct predictions (rows 1–3).** Both networks classify these three images correctly and look at the same things. In the tulip field (row 1) the baseline puts activation over the whole lower half and some of the upper rows, whereas A3's response is somewhat more compact over the centre of the lower flower bed; the rose (row 2) is covered by an almost identical red core over the bloom; for the sunflowers (row 3) both maps cover the central group of flower heads and neither activates the labels, bottles or white background in the lower half.
- **Background.** The background in these images is mostly dark or uniform, so there is little to suppress. The attention model does not show a visible, consistent increase in suppression of leaves or background.
- **Misclassified bouquet (row 4, true tulips, both predict roses).** Both models concentrate on the pink cluster at the lower centre of the bouquet; the upper, white-and-purple tulips receive little activation, and the model apparently reads the pink blossoms as roses.
- **Misclassified fence (row 5, true roses, both predict tulips).** The baseline's activation runs along the base of the rose bushes and the fence; A3's moves to the right-hand red roses. The prediction is unchanged, so the changed focus did not change the decision.

Quantification over all 550 test images. Two numbers summarise each Grad-CAM: *coverage* (the fraction of the image where the map exceeds 50% of its maximum; lower means more compact) and the *centre-box share* (the fraction of the map's total mass in the central 50% × 50% box; a uniform map gives 0.25).

Model	Coverage, all	Coverage, correct	Coverage, wrong	Centre share, all	Centre share, correct	Centre share, wrong	n wrong
A0 Baseline	0.245	0.251	0.153	0.459	0.463	0.383	30
A1 Channel (SE)	0.238	0.243	0.130	0.459	0.463	0.377	26
A2 Spatial	0.218	0.223	0.128	0.453	0.456	0.410	31
A3 C→S	0.227	0.235	0.098	0.455	0.460	0.359	31
A4 Self-Attention	0.288	0.296	0.177	0.408	0.411	0.363	36
A5 Combined	0.277	0.292	0.118	0.373	0.376	0.342	48
ECA	0.243	0.250	0.143	0.452	0.456	0.380	34
CBAM	0.214	0.222	0.100	0.480	0.484	0.414	36
Coordinate	0.321	0.328	0.214	0.422	0.425	0.387	37

- **Compactness.** The maps of A2, A3, CBAM and SE cover 21–24% of the image, somewhat less than the baseline's 24.5%; those of self-attention (28.8%), the combined model (27.7%) and Coordinate Attention (32.1%) are more spread out.
- **Centre bias.** The share of activation in the central box is 0.45–0.48 for the baseline, SE, spatial, A3, ECA and CBAM (nearly twice the 0.25 expected from a uniform map, as the flower is usually near the centre); it is lower for self-attention (0.41), Coordinate (0.42) and the combined model (0.37), whose activation is less centred.
- **Correct versus wrong predictions.** For every model the Grad-CAM of *wrongly* classified images is much more compact (coverage 0.10–0.21, against 0.22–0.33 for correct ones) and slightly less centred (0.34–0.41 versus 0.38–0.48). Wrong predictions are therefore made from a small, localized patch of evidence rather than a broad view of the flower.
- **Caveat.** None of these differences is related to accuracy: CBAM and A3 have the most compact maps and 36 and 31 errors; the baseline, with a broader map, has 30. Compactness is a property of where the model looks, not a measure of whether it is right.

28. Internal Attention Maps

An important objective of this experiment is to distinguish between

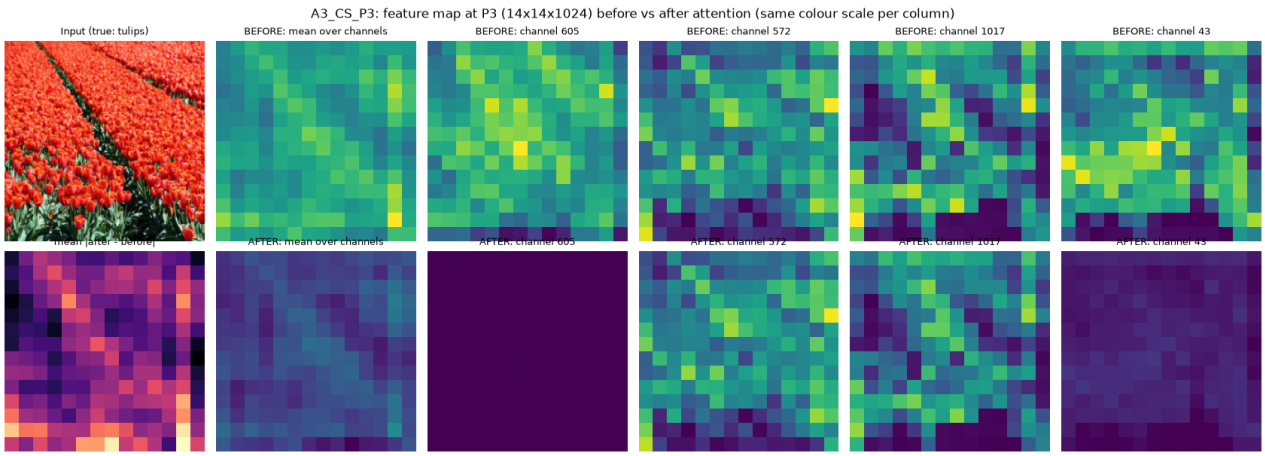
Internal Attention and Post-hoc Explanation

because they are not the same. An attention module directly computes weights during the forward pass and these weights multiply the features; Grad-CAM, in contrast, is generated after prediction using gradients and does not change the prediction. The following subsections show the weights the modules actually computed.

28.1 Channel-Attention Visualization

For channel attention, $M_c(F) = [m_1, m_2, \ldots, m_C]$. The learned channel weights of A3 at P_3 ($C = 1024$) were visualised: Plot 12 shows the feature map before and after the attention block for one example image, and Plot 13 the learned channel weights (top-20 channels and the sorted weights).

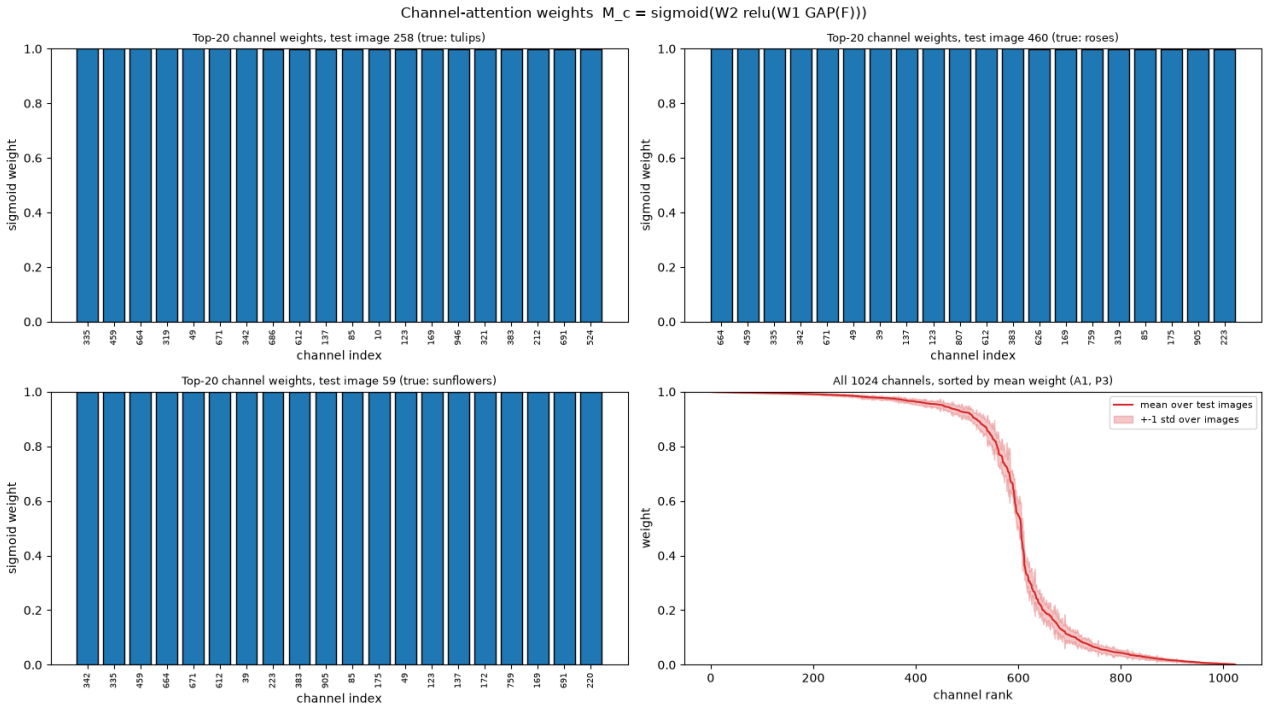
Plot 12: Feature Maps before and after Attention (A3 at P_3, $14 \times 14 \times 1024$, test image 258)



Inference:

- The figure shows individual channels before and after the block. Channel 605 is *completely zeroed* after attention and channel 43 is strongly attenuated, whereas channels 572 and 1017 pass almost unchanged. The mean feature map after attention is dimmer overall but keeps the same spatial pattern: the channel gate rescales *what* is passed on and leaves *where* intact.
- The bottom-left panel (mean |after − before|) is brightest along the bottom rows, which is where the spatial gate of this module has its low-value strip (Plot 14b), and weaker in the interior: the change made by the spatial gate is confined to the image border. A channel gate multiplies a whole channel by one number and cannot move activation in space.

Plot 13: Channel Attention Weights (top-20 channels and sorted weights, A3 at P_3)



Inference:

- **A small number of channels with strong weights?** The 20 highest weights are all close to 1.0, but they are not special: the sorted curve is flat near 1 for roughly the first 600 channels, then falls steeply and reaches almost 0 by rank 900–1000. In A3 61% (and in SE 59%) of the channels have weights above 0.5, and the top 10% of channels hold only 16.5% of the total weight mass — the gate is a *switch* that turns off roughly 40% of the channels, not a ranking of a few important ones.
- **Many channels contributing similarly?** Yes, the roughly 600 channels at the top all pass with weights near 1.
- **Does weighting change between images?** Almost not at all. The standard deviation of a channel's weight across images is 0.037 for A3 (and 0.013 for SE, A1), against 0.39 (A3) and 0.44 (A1) across channels, and the weight vectors of different images have a correlation of 0.981 (A3) and 0.998 (A1). The learned gate is a nearly fixed per-channel filter, closer to a learned re-weighting of the feature set than to input-dependent attention.

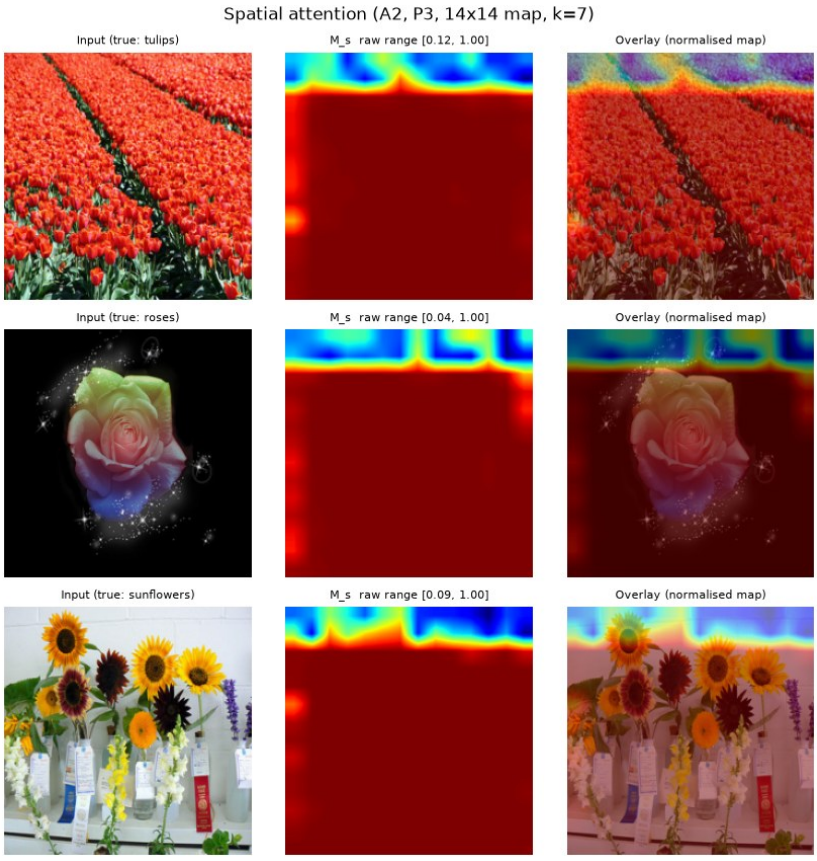
Channel-weight statistics for the other models that contain a channel gate (mean, across-channel std, across-image std, fraction above 0.5, top-10% share, between-image correlation):

Model	Mean weight	Std across channels	Std across images	Fraction > 0.5	Top-10% share	Corr. between images
A1 SE	0.590	0.438	0.0134	0.590	0.168	0.998
A3 C→S	0.599	0.390	0.0369	0.611	0.165	0.981
O1 S→C	0.603	0.404	0.0274	0.611	0.164	0.990
A5 Combined	0.552	0.322	0.0468	0.574	0.175	0.969
ECA	0.527	0.077	0.0552	0.620	0.129	0.437
CBAM	0.564	0.490	0.0079	0.563	0.177	0.994

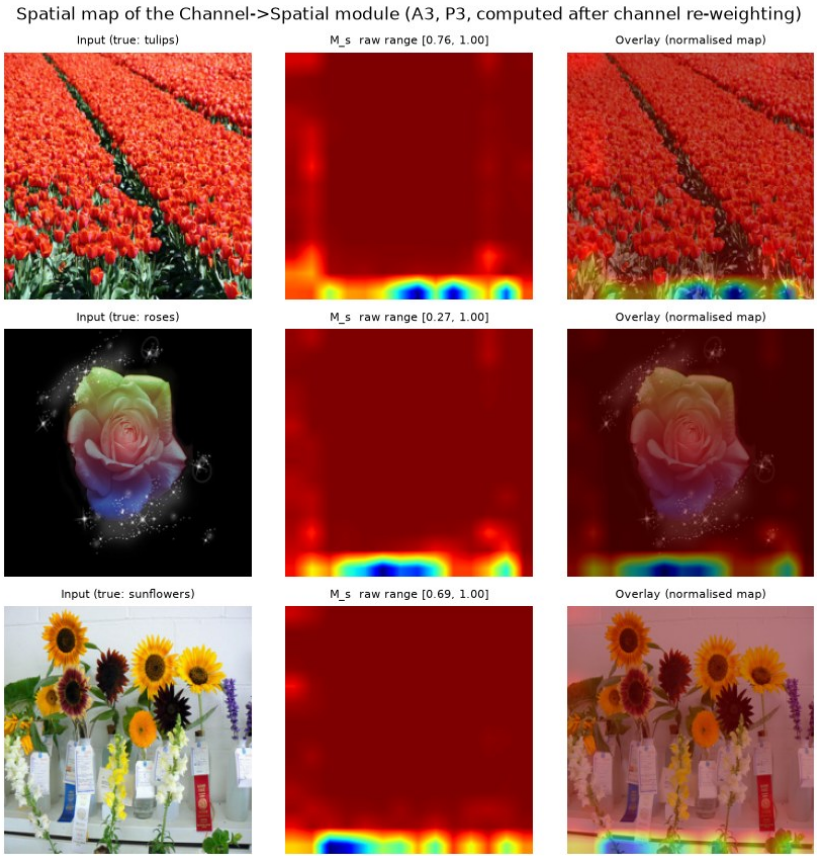
SE, CBAM and the C→S/S→C models are near-static switches (between-image correlation 0.98–0.99). ECA, whose 5-parameter convolution only sees neighbouring channels, is the opposite: the weights hardly differ between channels (std across channels only 0.077) and the pattern differs between images (correlation 0.44) — a mild, input-dependent modulation with little selective power.

28.2 Spatial-Attention Visualization

Plot 14a: Spatial Attention of A2 at P_3 (Input | Attention Map | Overlay)



Plot 14b: Spatial Map of the Channel→Spatial Module (A3 at P_3)



Inference:

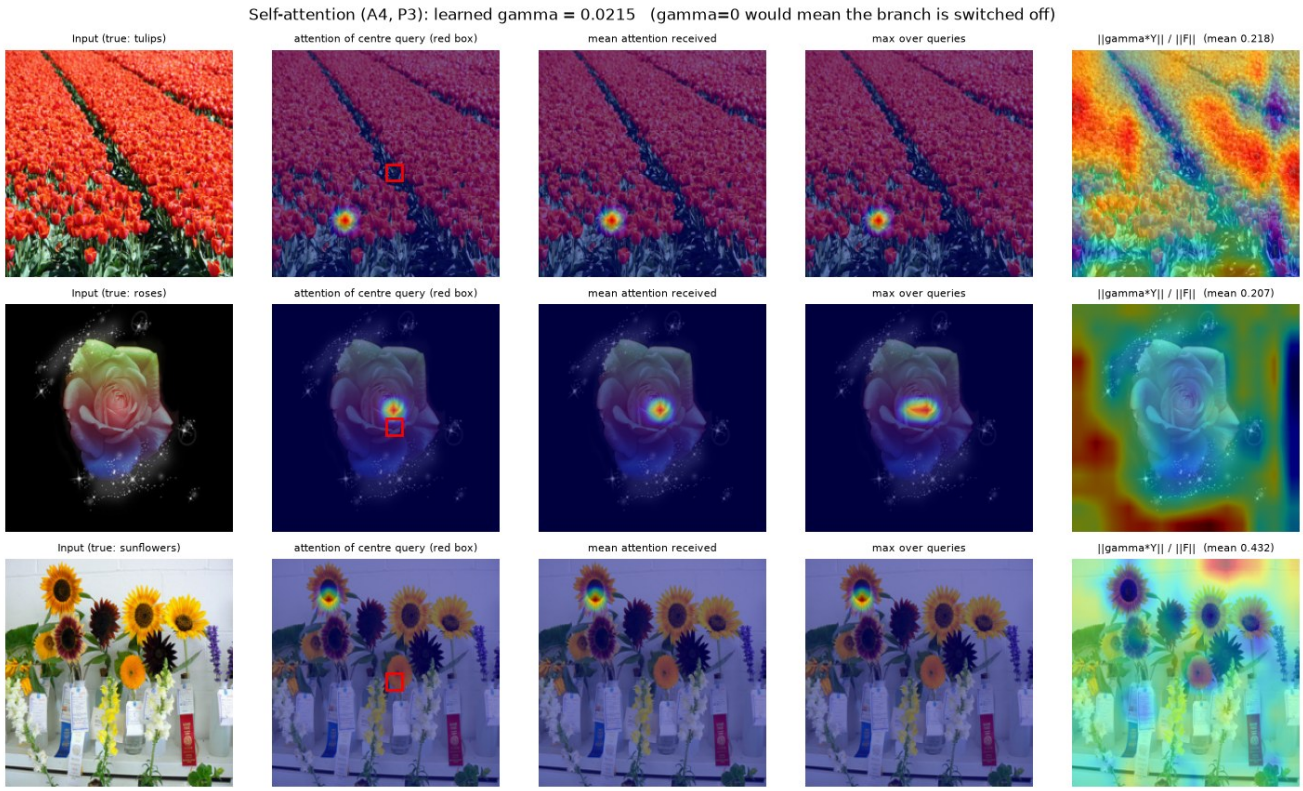
- **Which regions receive high attention?** *Almost all of the image.* The interior of every map is ≈ 1.0 (dark red) irrespective of image content: the same value over the flowers, the black background around the rose and the white wall. Only a thin strip at one image border has low values. For A2 this strip is at the **top** (raw minimum 0.04–0.12; the three images shown), whereas for A3 it is at the **bottom** (raw minimum 0.27–0.76).
- **Interpretation.** These maps do not outline objects. The strip lies at the image border for all five test images regardless of where the flower is, so it is most likely a border (zero-padding) effect — the 7×7 convolution sees zeros outside the 14×14 grid and can learn to give border cells a different value — and not selection of an object. The learned spatial attention is thus an almost constant gate with a border correction, which explains why it neither helped nor hurt accuracy (A2 and A3: -1 image).
- **Gate statistics** (mean raw value over all test images and cells; contrast is the mean over images of max minus min of the raw gate):

Spatial gate	Mean raw value	Contrast	Normalised entropy	Coverage (≥50% of max)	Centre-box share	Corr. between images
A2 Spatial, P_3	0.903	0.920	0.9914	0.885	0.277	0.897
A3 C→S, P_3 ($k = 7$)	0.985	0.374	0.9997	0.974	0.254	0.690
O1 S→C, P_3	0.937	0.852	0.9959	0.931	0.267	0.838
K3 ($k = 3$)	0.742	0.917	0.9904	0.883	0.262	0.650
K5 ($k = 5$)	0.974	0.563	0.9992	0.966	0.256	0.764
CBAM, P_3	0.987	0.344	0.9998	0.977	0.253	0.644
B1 C→S, P_1	0.978	0.725	0.9996	0.985	0.254	0.899
B2 C→S, P_2	0.968	0.875	0.9985	0.971	0.258	0.863
B4 C→S, P_4	0.995	0.152	0.9996	0.968	0.328	0.303

All gates have a normalised entropy above 0.99 (1.0 = perfectly uniform), 88–99% of the image is at least half as high as the maximum, and the centre-box share is 0.25–0.28 (0.25 is uniform; 0.33 for the 7×7 gate at P_4): none of them concentrates attention on the object or the centre. The sharpest gates (smallest mean value) belong to the 3×3 kernel (mean 0.742) and to the spatial-only gate A2 (0.903); $k = 3$ is also the weakest model of the kernel experiment (93.27%), while A2 is 0.18 pp below the baseline.

28.3 Self-Attention Visualization

Plot 15: Self-Attention of A4 at P_3 (query, mean received, maximum over queries, contribution)



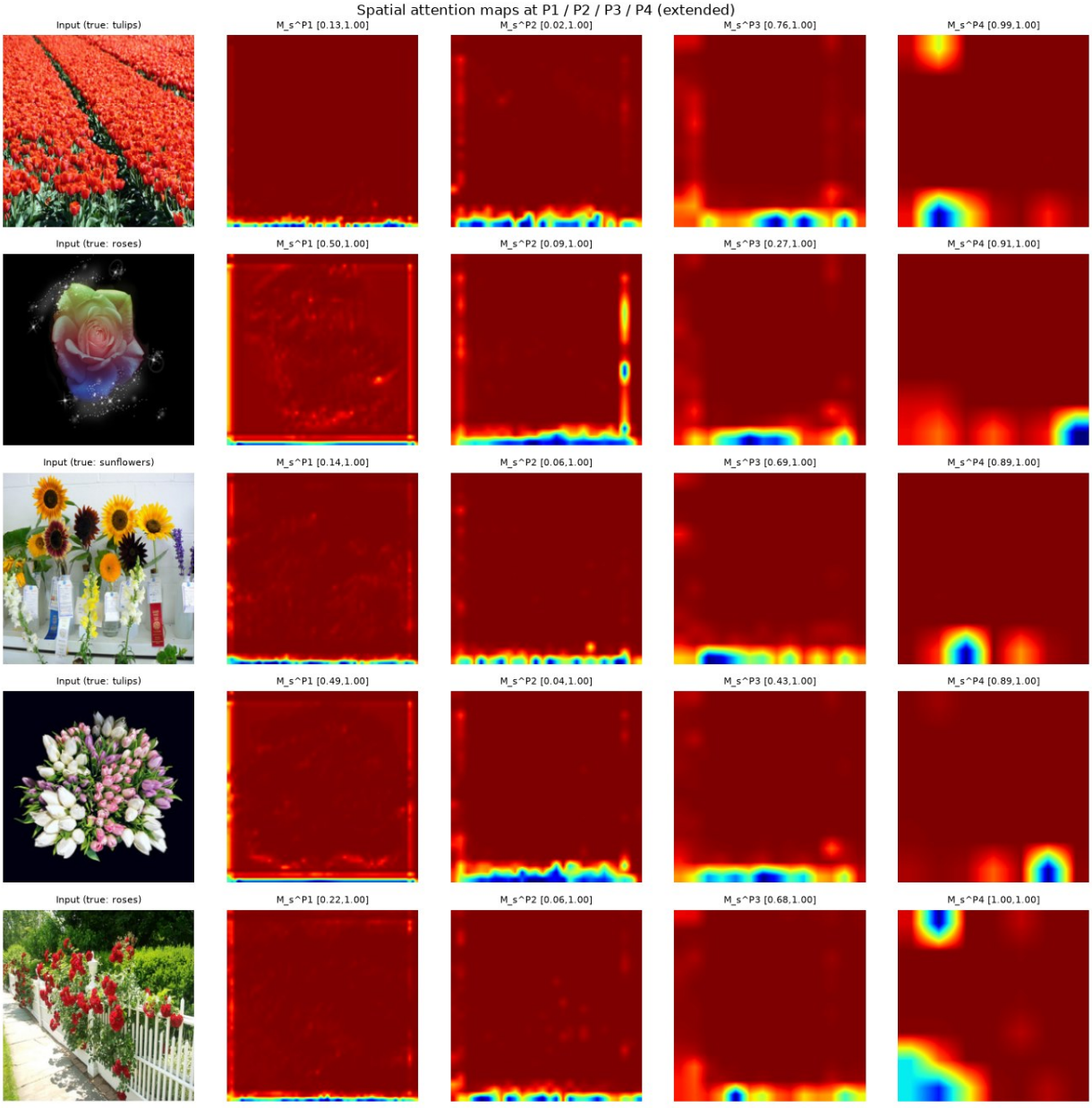
Inference: the columns show (2) the attention of the centre query (red box), (3) the mean attention each location receives, (4) the maximum over all queries, and (5) the relative size of the self-attention contribution $\|\gamma Y\|/\|F\|$ at every location.

- **How do distant regions interact?** Hardly selectively: in all three images the centre query, the mean-received map and the max-over-queries map all show the *same single hot spot* (for example the lower-left tulips, the centre of the rose, a sunflower head at the top). The attention collapses onto one location per image, which every query looks at; the attention is not a distribution over many related regions. The row entropy of the attention matrix is just 0.012 (A4) and 0.056 (A5).
- **Distance.** The mean attended distance is 6.3 grid cells (6.4 for A5), almost the 7.3 expected from uniform random attention on a 14×14 grid, and only 6.6% of the attention falls within 2 cells of the query (uniform: 5.9%): there is no locality bias.
- **Size of the contribution.** The learned scale is small ($\gamma = 0.0215$ for A4, -0.0210 for A5) but not negligible: the contribution γY has about 28% (A4) and 30% (A5) of the norm of the input features F; it is 0.21–0.22 for the tulip and rose images and 0.43 for the sunflower image. In the rose and sunflower images the contribution is smallest over the flower and largest in the surroundings. In A5, γ is even negative.
- **Conclusion.** Self-attention here behaves as a learned global bias term rather than as a mechanism that links related regions, which agrees with its failure to improve accuracy (93.45%, -1.09 pp) despite 1.31 million parameters.

29. Attention Position Visualization

The spatial attention maps learned at different ResNet50 stages (modules B1, B2, A3 and B4) are shown for the same five test images; P_1 was added to the three positions of the manual.

Plot 16: Spatial Attention Maps at P_1, P_2, P_3 and P_4 (the same five test images)



Inference:

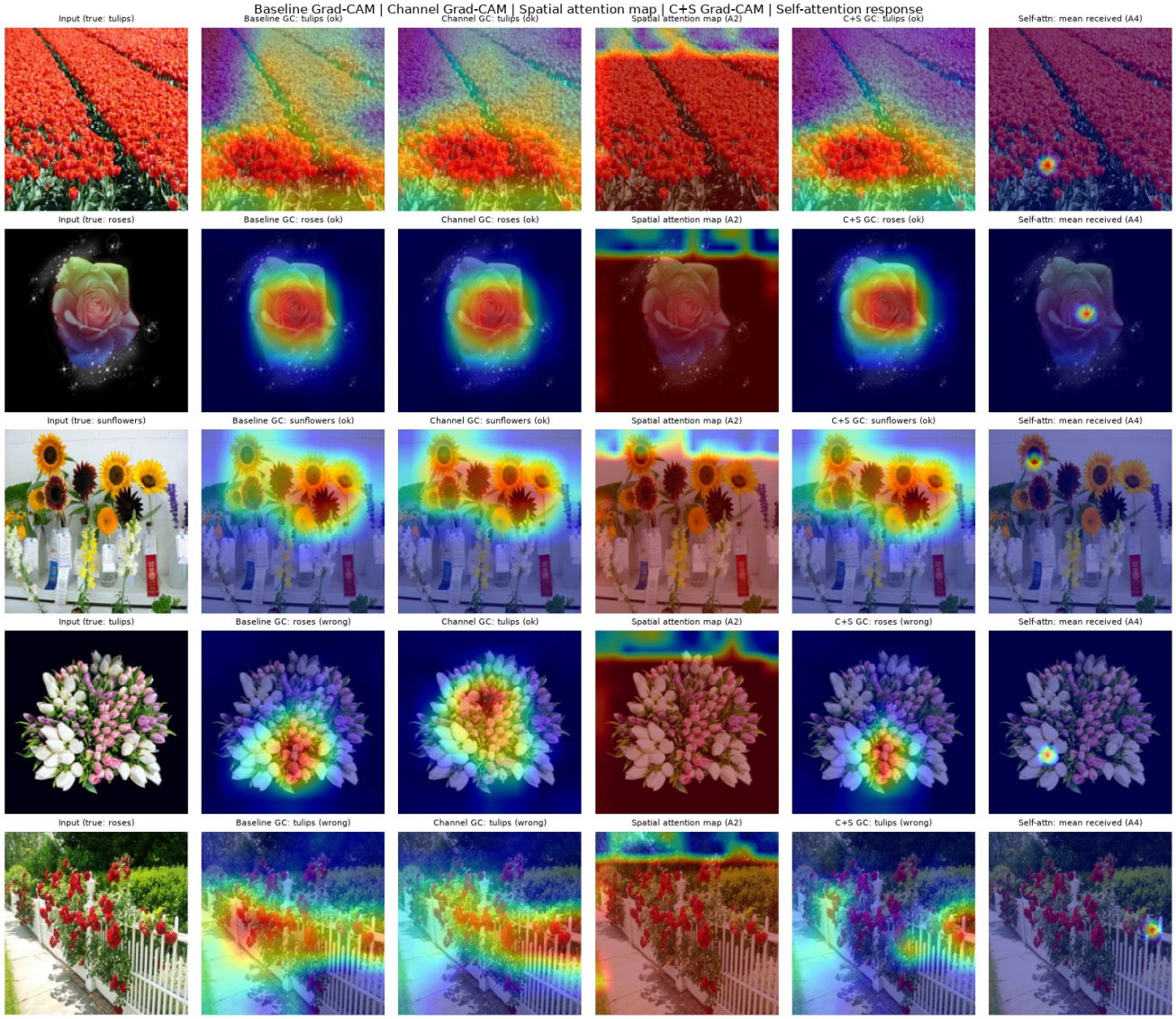
- P_1 (56×56): almost the entire map is at 1.0; the minimum is a thin frame around the image border (raw minimum 0.13–0.50, mean raw value 0.978).
- P_2 (28×28): a strong, thicker strip along the bottom edge (raw minimum 0.02–0.09) and, for the rose image, also the right edge. The interior is again ≈ 1.0.
- P_3 (14×14): the bottom strip is wider but shallower (raw minimum 0.27–0.76).
- P_4 (7×7): the map is one to two coarse corner cells low (raw minimum 0.89–1.00) at different corners depending on the image; the gate is essentially 1.0 everywhere (mean raw value 0.995, contrast 0.152).

The maps do *not* show the hoped-for progression from edges and textures (P_2) through flower parts (P_3) to whole flower regions (P_4): at every stage the gate is ≈ 1 almost everywhere with a border artefact that is a thin line at P_1 and becomes coarser as the grid gets smaller (single corner cells at P_4). The mean gate contrast is 0.73 and 0.88 at P_1 and P_2 but only 0.37 and 0.15 at P_3 and P_4, and the correlation between images is 0.90, 0.86, 0.69 and 0.30 for P_1–P_4 (P_4 is the most image dependent, but it has only 49 cells). Spatial attention therefore does little in all positions in this run, and the earliest-position accuracy loss (B1, -1.64 pp) cannot be explained by a visible object-suppression pattern; the more likely cause is the down-scaling of low-level features (Section 15).

30. Qualitative Comparison Across Attention Mechanisms

Three correctly classified images (test indices 258, 460, 59) and two incorrectly classified images (19 and 334) were compared across models, in the arrangement the manual recommends: input, baseline Grad-CAM, channel-attention (SE) Grad-CAM, the spatial-attention map (A2), Channel+Spatial Grad-CAM, and the self-attention response (A4).

Plot 18: Qualitative Comparison of Baseline, Channel, Spatial, Channel+Spatial and Self-Attention Responses



30.1 Qualitative Criteria

Inference (all statements are about what is visible in Plots 17, 18 and 19):

1. **Foreground localization.** On the single rose (row 2) all three Grad-CAMs form a red core over the bloom with a blue (inactive) background. On the sunflower image (row 3) they cover the group of heads in the upper centre. The baseline and the attention models localize the foreground equally well.
2. **Background suppression.** In the images with a black background the background is already inactive for every model; in the tulip field (row 1) the Channel+Spatial map is a little more compact than the baseline's, with a less active upper half, while the SE map is almost the same as the baseline's; all maps still extend over a large part of the bed.
3. **Petals and flower centre.** At 7×7 resolution a map cannot resolve petals; on the rose the maximum is over the centre of the bloom for every model, and on the sunflower heads the maximum is on the flower discs.
4. **Compactness and coverage.** See the table in Section 27.1: A3, CBAM and spatial attention are the most compact; self-attention, the combined model and Coordinate Attention the most spread out.
5. **Incorrectly attended regions.** For the misclassified fence image (row 5) the baseline's evidence lies along the base of the bushes and fence and A3 moves to the right-hand red roses — both predict tulips. The six models of Plot 19, and the other three Grad-CAM models (A2, A3, A5), also predict tulips for this image, so the error does not depend on the mechanism: the red flowers against a white fence and green hedge are consistently read as tulips.
6. **Correct versus incorrect samples.** Wrong predictions have much more compact maps than correct ones for every model (Section 27.1).
7. **Shallow versus deep attention.** The spatial maps of the shallow stages show a thin frame (P_1) or border strip (P_2); the deep stage shows one or two coarse cells (Plot 16). No stage shows object-shaped attention.
8. **Consistency across images.** The channel gate is almost identical for all images (correlation 0.98–0.998), spatial gates have a fixed border pattern (correlation 0.64–0.90 for P_1–P_3 models), and self-attention picks one different hot spot per image.
9. **Differences between correct and wrong images, row 4 (true tulips, white-and-pink bouquet).** Here the mechanisms *do* disagree: the baseline, spatial attention, Channel+Spatial, ECA and self-attention predict roses, while SE, the combined model, CBAM and Coordinate Attention predict tulips. The three correct models spread activation over the upper centre of the bouquet, as well as the pink cluster at the lower centre that the wrong ones use. This is the one example in the five where a change in evidence coincides with a change in decision; with one image it is an anecdote, not evidence that these mechanisms localize better in general.

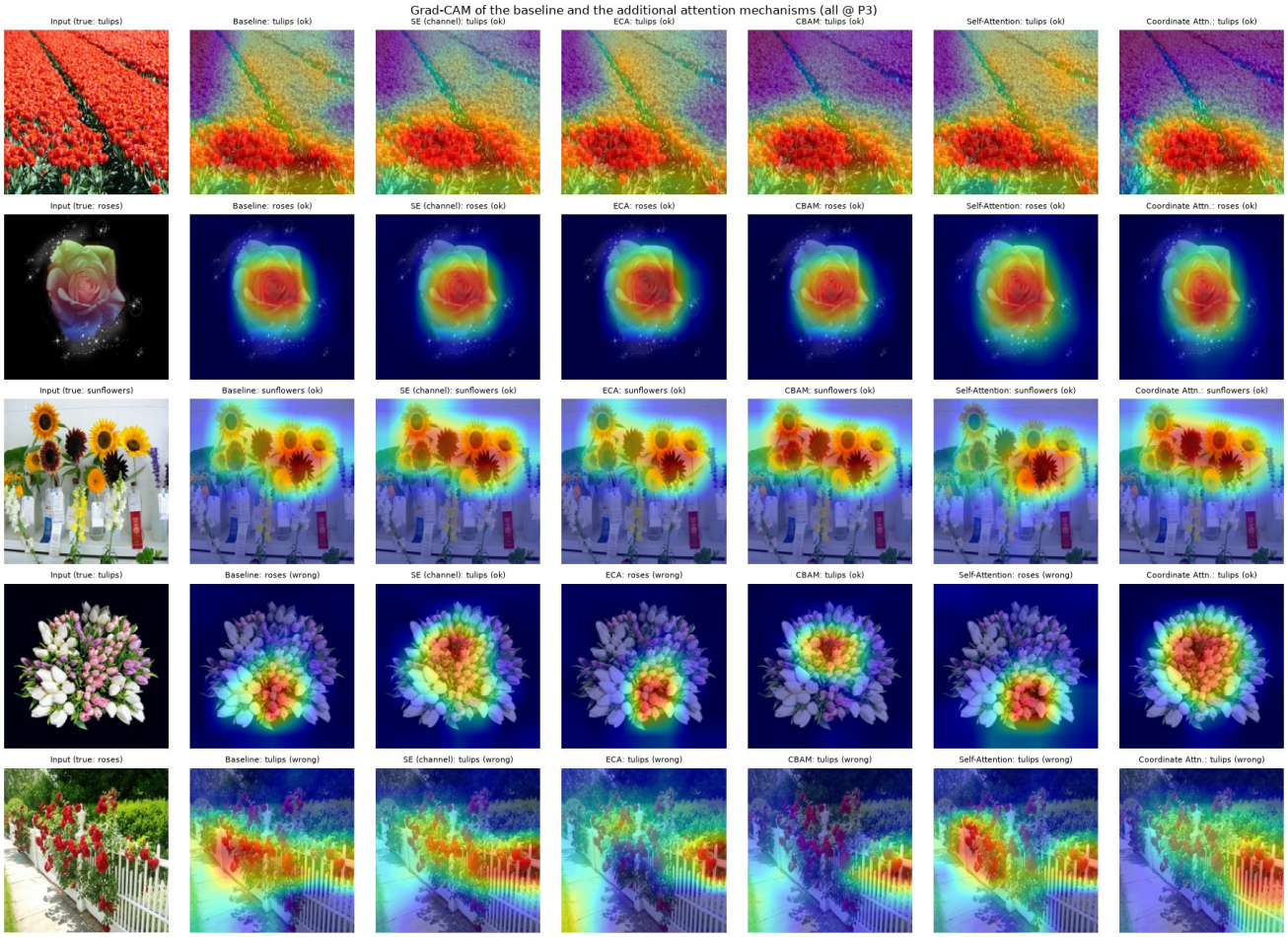
31. Required Visualizations

No.	Required visualization	Where in this report
1	sample input images from all five classes	Plot 1 (Section 3.2)
2	training and validation accuracy curves	Plot 2, Plot 4 (Sections 24.1, 33)
3	training and validation loss curves	Plot 3, Plot 4 (Section 24.2)
4	confusion matrix	Plot 11 (Section 25)
5	feature map before attention	Plot 12, top row (Section 28.1)
6	channel-attention weights	Plot 13 (Section 28.1)
7	spatial-attention heatmap	Plot 14a, 14b (Section 28.2)
8	spatial-attention overlay	Plot 14a, 14b (third column)
9	self-attention response	Plot 15 (Section 28.3)
10	feature map after attention	Plot 12, bottom row (Section 28.1)
11	baseline Grad-CAM	Plots 17, 18, 19
12	Grad-CAM of at least one attention model	Plots 17, 18, 19
13	attention maps at P_2, P_3, P_4	Plot 16 (Section 29)
14	graphical comparison of attention mechanisms	Plots 5–10 (Section 24)

32. Recommended Visualization Layout

The layout of the manual (input | baseline | channel | spatial | channel+spatial) is shown in Plot 18 (Section 30), where the models are arranged horizontally for five images. The extended layout with the additional mechanisms is shown in Plot 19: the baseline, SE, ECA, CBAM, self-attention and Coordinate Attention for the same five images.

Plot 19: Grad-CAM of the Baseline and the Additional Attention Mechanisms (all at P_3)



Inference:

- **Correct images (rows 1–3):** all six models predict correctly and produce similar maps: a broad response on the lower tulip bed, a red core over the rose and over the central sunflowers. The differences between mechanisms are differences in the width of the region, not in its location (the quantitative coverage differences are in the table of Section 27.1).
- **Row 4 (bouquet):** SE, CBAM and Coordinate Attention are correct and ECA, the baseline and self-attention are wrong (see Section 30.1, item 9).
- **Row 5 (rose fence):** none of the six models shown gets this image right. Their maps differ — self-attention covers the left-hand bushes, Coordinate Attention the fence and right-hand roses, ECA and CBAM the right side — but all end up in the same wrong class, so *different evidence leads to the same error*.
- **Conclusion.** On these five images the mechanisms change where the Grad-CAM peaks, but only one of the five predictions depends on the mechanism.

33. Required Experimental Analysis

The analysis has three parts: quantitative, computational and qualitative.

33.1 Quantitative Analysis

$$\boxed{\text{Accuracy} + \text{Precision} + \text{Recall} + \text{Macro F1}}$$

The primary results are in Sections 14–19 and 23. Because each model was trained once and the test set has only 550 images, the table below adds a 95% bootstrap confidence interval (1000 paired resamples of the test set) and the

exact McNemar test against the baseline A0. "Only A0 correct" and "Only model correct" count the test images that exactly one of the two models classifies correctly.

Model	Acc. (%)	95% CI	Macro F1 (%)	95% CI	Δ vs A0 (pp)	Only A0 correct	Only model correct	McNemar p
A0: Baseline	94.55	[92.4, 96.4]	94.59	[92.5, 96.3]	–	–	–	–
A1: Channel (SE)	95.27	[93.3, 96.9]	95.17	[93.2, 96.9]	+0.73	6	10	0.454
A2: Spatial (7×7)	94.36	[92.4, 96.2]	94.35	[92.4, 96.2]	-0.18	6	5	1.000
A3: Channel→Spatial	94.36	[92.4, 96.4]	94.35	[92.2, 96.3]	-0.18	9	8	1.000
A4: Self-Attention	93.45	[91.5, 95.3]	93.42	[91.3, 95.3]	-1.09	11	5	0.210
A5: C→S→Self	91.27	[88.9, 93.6]	91.19	[89.0, 93.5]	-3.27	29	11	0.006
B1: C→S at P_1	92.91	[90.5, 95.1]	92.92	[90.5, 95.1]	-1.64	16	7	0.093
B2: C→S at P_2	94.73	[92.7, 96.4]	94.70	[92.7, 96.4]	+0.18	9	10	1.000
B4: C→S at P_4	94.55	[92.5, 96.4]	94.47	[92.4, 96.3]	+0.00	7	7	1.000
M34: C→S at P_3+P_4	94.36	[92.4, 96.2]	94.29	[92.3, 96.2]	-0.18	10	9	1.000
M234: C→S at P_2+P_3+P_4	93.64	[91.3, 95.6]	93.64	[91.2, 95.6]	-0.91	13	8	0.383
O1: Spatial→Channel	94.91	[92.9, 96.7]	94.87	[92.9, 96.6]	+0.36	7	9	0.804
K3: C→S, $k = 3$	93.27	[91.1, 95.5]	93.15	[90.9, 95.2]	-1.27	14	7	0.189
K5: C→S, $k = 5$	94.18	[92.2, 96.2]	94.11	[92.1, 96.1]	-0.36	11	9	0.824
X1: ECA	93.82	[91.6, 95.5]	93.74	[91.6, 95.5]	-0.73	9	5	0.424
X2: CBAM	93.45	[91.5, 95.5]	93.44	[91.3, 95.4]	-1.09	14	8	0.286
X3: Coordinate Attention	93.27	[91.1, 95.5]	93.10	[90.8, 95.3]	-1.27	17	10	0.248

Inference:

- **Intervals are about 4 pp wide.** Every accuracy interval spans roughly ±2 pp and they overlap almost completely: the baseline's is 92.4–96.4%, and SE's, the best, is 93.3–96.9%.
- **Only one difference is distinguishable.** A5 (C→S→Self) is worse than the baseline ($p = 0.006$; the baseline is right and A5 wrong on 29 images, the reverse on 11). The next smallest p-values are B1 (0.093, attention at P_1), K3 (0.189), A4 (0.210), X3 (0.248) and X2 (0.286); none is below 0.05.
- **SE's advantage is small.** SE fixes 10 baseline errors and breaks 6 ($p = 0.454$), a net gain of 4 images. Spatial→Channel (O1) fixes 9 and breaks 7 ($p = 0.804$).
- **What this test set can detect.** With a baseline error of about 5.5% (30 images), differences smaller than about 2 pp (about 11 images) cannot be told apart from chance, so the study can rule out large effects but cannot rank the many models within 2 pp of the baseline. In particular the order C→S versus S→C ($p = 0.508$), the position P_2 versus P_3 versus P_4, and the kernel sizes $k = 5$ versus $k = 7$ are all within noise.
- **Precision, recall and macro F1** agree with accuracy for every model (Sections 14 and 19): they differ from accuracy by at most 0.3 pp, because the classes are moderately balanced and no model trades precision against recall in a systematic way.

33.2 Complexity Analysis

Parameters + Inference Time

Model	Attention params	Total params	Trainable (phase 1)	Trainable (phase 2)	Added vs A0 (%)	ms/img (bs 32)	Time vs A0 (%)	Train (min)
A0: Baseline	0	23,597,957	10,245	14,963,717	+0.00	1.18	+0.0	14.3
A1: Channel (SE)	132,160	23,730,117	142,405	15,095,877	+0.56	1.21	+2.8	14.1
A2: Spatial (7×7)	99	23,598,056	10,344	14,963,816	+0.00	1.21	+2.8	14.0
A3: Channel→Spatial	132,259	23,730,216	142,504	15,095,976	+0.56	1.22	+3.4	14.0
A4: Self-Attention	1,310,721	24,908,678	1,320,966	16,274,438	+5.55	1.27	+7.2	14.5
A5: C→S→Self	1,442,980	25,040,937	1,453,225	16,406,697	+6.11	1.33	+12.7	14.5
B1: C→S at P_1	8,563	23,606,520	18,808	14,972,280	+0.04	1.26	+6.7	14.2
B2: C→S at P_2	33,411	23,631,368	43,656	14,997,128	+0.14	1.24	+4.6	14.2
B4: C→S at P_4	526,563	24,124,520	536,808	15,490,280	+2.23	1.22	+3.0	14.2
M34: C→S at P_3+P_4	658,822	24,256,779	669,067	15,622,539	+2.79	1.23	+3.9	14.1
M234: C→S at P_2+P_3+P_4	692,233	24,290,190	702,478	15,655,950	+2.93	1.25	+6.2	14.3
O1: Spatial→Channel	132,259	23,730,216	142,504	15,095,976	+0.56	1.22	+2.9	14.0
K3: C→S, $k = 3$	132,179	23,730,136	142,424	15,095,896	+0.56	1.22	+3.4	14.1
K5: C→S, $k = 5$	132,211	23,730,168	142,456	15,095,928	+0.56	1.22	+3.4	14.0
X1: ECA	5	23,597,962	10,250	14,963,722	+0.00	1.22	+2.9	14.3
X2: CBAM	132,259	23,730,216	142,504	15,095,976	+0.56	1.23	+3.7	14.2
X3: Coordinate Attention	100,512	23,698,469	110,693	15,064,165	+0.43	1.23	+3.7	14.1

"Added vs A0" is the relative increase of the total parameter count; "Time vs A0" is the relative increase of the batch-32 inference time. Single-image latency (batch size 1) was also recorded by the notebook but is not used, since it varies from 9.2 to 19.2 ms with no relation to the architecture (Section 22). Training time includes validation after every epoch.

Inference:

- **Parameters.** The total parameter count ranges from 23.60 M (baseline, ECA, spatial) to 25.04 M (A5, +6.1%). Only the self-attention models (A4, A5), the P_4 module (B4, 527 thousand parameters, +2.2%) and the multi-position models (M34, M234, +2.8 to +2.9%) add more than 1%. During fine-tuning the baseline updates 14.96 M parameters (conv5_x and the head); the additional modules enlarge that by at most 10% (A5: 16.41 M).

- **Inference time.** The baseline needs 1.18 ms/img; all modules cost +2.8% to +7.2% except the combined model (+12.7%). Time does not scale with parameters: the 5-parameter ECA costs +2.9% and the 132 K-parameter SE module +2.8%, whereas P_1 attention with only 8.6 K parameters costs +6.7% because its feature map has 56×56 locations.
- **Training time.** Every model takes 14.0–14.5 minutes for 15 epochs (self-attention models the longest at 14.5 min): the pretrained backbone dominates.
- **Memory.** The self-attention matrix needs 0.15 MB per image at P_3 but would need 37.5 MB per image at P_1 (Section 13); this is why self-attention was not trained at P_1.
- **Return on cost.** SE improves accuracy by 0.73 pp for +0.56% parameters and +2.8% time; the two self-attention models cost +5.6–6.1% parameters and +7–13% time and lose 1.1–3.3 pp.

33.3 Qualitative Analysis

Grad-CAM + Attention Maps + Visual Interpretation

The qualitative analysis (Sections 26–32) shows the following.

- **Grad-CAM.** The baseline and the attention models look at the same flower; the mechanisms change the width of the response (spatial attention, CBAM and A3 more compact, self-attention and Coordinate Attention more spread out) but not its location, and misclassified images have more compact maps for every model.
- **Internal attention maps.** The channel gate is a nearly fixed switch (correlation between images 0.98–0.998) that turns off about 40% of the channels; the spatial gates are ≈ 1 everywhere except at a border strip; self-attention collapses onto one location per image.
- **Visual interpretation.** None of the learned maps outlines the flower or suppresses the background selectively, so the attention modules in this run act as re-weighting layers, not as object selectors. The quantitative result — no measurable gain — is consistent with this.

Overall evaluation: **quantitative** (no reliable difference in accuracy or F1 from the baseline except a loss for the combined model) + **computational** (cheap modules cost 3–4% time, expensive ones 7–13%) + **qualitative** (attention maps are nearly constant and do not focus on the object).

34. Experimental Protocol

To ensure a fair comparison all 17 models used (every item identical, verified in the notebook code):

- the same training, validation and test images (one stratified split, MD5 fingerprints `f7971466`, `2dddaabb`, `dabcfc07`);
- the same random seed (42), reset before every model, with a freshly created augmentation pipeline and the same shuffling seed;
- the same preprocessing (`resnet50.preprocess_input`) and augmentation (flip, rotation, zoom, translation, contrast);
- the same backbone (ImageNet ResNet50, re-assembled with exactly the same output as the original, max. difference 0.0) and the same classifier (GAP, dropout 0.3, Dense-5);
- the same optimizer (Adam), batch size (32) and number of epochs (5 frozen + 10 fine-tuning = 15, no early stopping);
- the same learning-rate schedule (10^{-3} constant, then 10^{-4} with cosine decay) and the same fine-tuning procedure (conv5_x unfrozen, batch-normalisation frozen);
- the same rule for selecting the epoch (best validation accuracy) and the same evaluation (the same test images and the same metric code).

Only the factor under investigation changed in each experiment: the attention type in Experiment 1 (position P_3), the position in Experiment 2 (module C→S), the number of positions in Experiment 3, the order in Experiment 4 and the kernel size in Experiment 5.

Remaining sources of variation. Each model was trained once. GPU operations were not forced to be deterministic, so rerunning a model may flip a few predictions; and a module that adds parameters changes the random-number stream used for initialisation in later layers. The differences below about 2 pp in this report must therefore be regarded as within the run-to-run noise. Section 40 discusses this limitation.

35. Minimum Mandatory Ablation Set

All ten mandatory items were completed:

No.	Required item	Model / figure	Result (test accuracy) / where
1	Baseline pretrained ResNet50	A0	94.55% (Section 14)
2	Channel attention at P_3	A1	95.27% (Section 14)
3	Spatial attention at P_3	A2	94.36% (Section 14)
4	Channel+Spatial at P_3	A3 (= B3)	94.36% (Sections 14, 15)
5	Channel+Spatial at P_2	B2	94.73% (Section 15)
6	Channel+Spatial at P_4	B4	94.55% (Section 15)
7	Self-attention at P_3	A4	93.45% (Section 14)
8	Grad-CAM, baseline versus one attention model	Plot 17	Section 27 (A3)
9	Spatial-attention visualization at P_2, P_3, P_4	Plot 16	Section 29
10	Graphical comparison of the classification performance	Plots 5–10	Section 24

Together these items allow the study of **attention type** (items 1–4, 7) and **attention position** (items 4–6). Beyond the minimum, the study also contains the extended items of Sections 16–19: attention at P_1, multiple positions, order, kernel size, the combined module and three additional mechanisms.

36. Additional Student Exploration

The additional exploration (Section 19) used **ECA**, **CBAM** and **Coordinate Attention** as extra mechanisms, together with SE (model A1) and lightweight self-attention (A4), all on the same ResNet50 backbone, with the same split, preprocessing, augmentation, optimizer, epochs, fine-tuning strategy and attention position (P_3).

- **Quantitative comparison:** Section 19.1 and 23 (accuracy, precision, recall, F1, parameters, time) and Section 33.1 (confidence intervals and tests). SE 95.27% > ECA 93.82% > CBAM = self-attention 93.45% > Coordinate 93.27%; none of the differences to the baseline is statistically distinguishable.
- **Qualitative comparison:** Plot 19 (Grad-CAM) and the statistics of Sections 27.1 and 28.1: ECA's channel gate is almost flat and image-dependent (std across channels 0.077, correlation between images 0.44), whereas SE and CBAM produce near-static switches (0.99); CBAM's spatial gate is almost uniform (mean 0.987, contrast 0.344); Coordinate Attention has the widest Grad-CAM (coverage 0.32).

37. Questions Students Must Answer

1. **What is the purpose of residual connections in ResNet50?** A residual block outputs $F(x) + x$, so the block only has to learn a correction to its input. The identity path lets gradients flow directly to earlier layers, which makes very deep networks (50 layers) trainable and easier to optimise, and it also means that a stage cannot do much worse than the identity.
2. **What does conv2_x represent compared with conv5_x?** conv2_x is the first stage after the stem: a 56×56 map with 256 channels, whose cells see a small neighbourhood and respond to edges, colours and simple textures. conv5_x is the last stage: a 7×7 map with 2048 channels, whose cells each summarise a 32×32-pixel area with a very large receptive field and respond to flower parts and the whole object.
3. **Why does the spatial resolution decrease with network depth?** Stride-2 convolutions (and the max-pool in the stem) halve the grid at each stage ($56 \to 28 \to 14 \to 7$). This enlarges the receptive field cheaply, gives some translation invariance and keeps the computation per stage roughly constant.
4. **Why does the number of feature channels increase with network depth?** As the spatial detail is reduced, the network stores more different feature detectors per location ($64 \to 256 \to 512 \to 1024 \to 2048$) because deeper features are combinations of earlier ones and higher-level concepts are more varied than edges.
5. **What is meant by low-level, intermediate and high-level features?** Low-level: edges, colours, orientations, textures (conv2_x). Intermediate: petals, repeated structures, part arrangements (conv3_x–conv4_x). High-level: object-level, class-discriminative information such as "this is a tulip" (conv5_x).
6. **What is the purpose of an attention mechanism?** To let the network re-weight its own features according to the input: emphasising useful channels, locations or relations, and suppressing the rest, at a small computational cost.
7. **What does channel attention attempt to learn?** A weight in $(0, 1)$ for every feature channel, computed from globally pooled information, i.e. "which channels are important" (Section 9). In this run the learned weights were nearly input-independent: about 59–61% of the channels of SE and A3 had weights above 0.5 and the weight vectors of different images correlated at 0.98–0.998 (Section 28.1).
8. **What does spatial attention attempt to learn?** A weight in $(0, 1)$ for every spatial location, computed from the channel-wise mean and max, i.e. "where is the important content". In this run the learned maps were ≈ 1 almost everywhere, with a low-value strip along one image border (Section 28.2).

9. **How is channel attention different from spatial attention?** Channel attention outputs a $1 \times 1 \times C$ vector (one number per channel, the same everywhere in the image) and has $\approx C^2/8$ parameters (132,160 at P_3); spatial attention outputs an $H \times W \times 1$ map (the same weight for all channels at a location) from a single small convolution (99 parameters).
10. **Why can channel and spatial attention be used sequentially?** They act on different axes, so one can be applied to the output of the other: $F_{cs} = M_s(M_c(F) \odot F) \odot (M_c(F) \odot F)$. Channel attention decides what, spatial attention where. The order may matter (spatial gates see the re-scaled features), but in this run C→S (94.36%) and S→C (94.91%) differed by only 3 images ($p = 0.51$).
11. **Why might channel attention behave differently at conv3_x and conv5_x?** At conv3_x (512 channels, 28×28) the channels encode generic textures and edges, which are similar across classes; at conv5_x (2048 channels, 7×7) they encode class-specific concepts, so gating them can directly change the evidence for a class. The module is also four times larger at conv5_x (526 K versus 33 K parameters). Here the effect on accuracy was small at both stages (B2 94.73%, B4 94.55%); the only clear difference was at the shallow end, where attention at P_1 lowered accuracy (92.91%).
12. **Why can spatial attention be useful at an intermediate ResNet stage?** At conv3_x–conv4_x the grid (28×28, 14×14) is still fine enough to localise flower parts, while the features already contain object-level information, so a spatial gate could in principle emphasise the flower and suppress the background. In this experiment, however, the learned spatial maps did not do this (Section 28.2), and A2/A3 at P_3 matched the baseline within one image.
13. **Why is self-attention computationally expensive when the feature map has high spatial resolution?** It computes an $N \times N$ similarity matrix between all $N = HW$ locations; cost and memory grow with N^2, not with N.
14. **What happens to the size of the self-attention matrix when the spatial resolution increases?** It grows with the fourth power of the side length: 2,401 entries at 7×7, 38,416 at 14×14 (16 times more), 614,656 at 28×28 and 9,834,496 at 56×56 (37.5 MB per image, about 1.2 GB for a batch of 32).
15. **How does attention performance vary when the insertion position is changed?** For C→S: 92.91% (P_1), 94.73% (P_2), 94.36% (P_3) and 94.55% (P_4) against 94.55% for the baseline: only P_1 is clearly lower (Section 15).
16. **How does attention at P_2 differ visually from attention at P_4?** The P_2 map (28×28) has a strong, narrow border strip (raw minimum 0.02–0.09; also the right edge for the rose image); the P_4 map (7×7) is ≈ 1 everywhere except one or two coarse corner cells (raw minimum 0.89–1.00) and has a much smaller contrast (0.15 versus 0.88). Neither outlines the flower (Plot 16).
17. **Does applying attention at multiple stages always improve classification performance?** No. P_3+P_4 (94.36%) equals P_3 alone and is below P_4 alone (94.55%); P_2+P_3+P_4 is lower still (93.64%) while adding 692 K parameters (Section 16).
18. **Does Channel→Spatial attention behave differently from Spatial→Channel attention?** The learned spatial gates do: S→C has a much more contrasted gate (mean 0.937, contrast 0.852) than C→S (0.985, 0.374). Accuracy was 94.91% versus 94.36% (3 images, $p = 0.51$), so the classification difference is not significant (Section 17).
19. **How does the spatial-attention kernel size influence the attention map?** The smaller the kernel, the more the gate departs from 1: mean 0.742 ($k = 3$), 0.974 ($k = 5$), 0.985 ($k = 7$) and contrast 0.917, 0.563, 0.374. A larger kernel gives a smoother, more uniform map (Section 18). Accuracy rose with k: 93.27%, 94.18%, 94.36%.
20. **How does SE attention differ from ECA attention?** SE uses two fully connected layers with a reduction ratio of 16 (132,160 parameters at $C = 1024$), so every channel can influence every other. ECA drops the dimension reduction and applies a 1-D convolution with $k = 5$ across neighbouring channels (5 parameters), so each channel interacts only with its 4 neighbours. Results: SE 95.27% vs. ECA 93.82%; ECA's gate varied little across channels (std 0.077) while SE's was a sharp switch (0.438), so ECA acts as a mild, input-dependent re-scaling while SE acts as a learned channel selection.
21. **How does CBAM combine channel and spatial attention?** Sequentially: the channel gate (a shared MLP applied to both average-pooled and max-pooled channel descriptors, summed before the sigmoid) first, then the 7×7 spatial gate computed from the channel-wise mean and max of the refined map. It has the same parameter count as the C→S module (132,259 at P_3) and, in this run, reached 93.45% (A3: 94.36%), a difference of 5 images that is not significant.
22. **How does self-attention differ from channel attention?** Channel attention re-scales whole channels using one global descriptor and does not mix information across space; self-attention computes pairwise relations between all locations and adds a weighted sum of values from other places (a residual with a learned scale γ). Self-attention has 10× more parameters (1.31 M versus 132 K) and its cost grows with N^2. In this run it collapsed onto one location per image (Section 28.3) and lowered accuracy by 1.09 pp.
23. **Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?** No. The two largest modules (A5, 1.44 M parameters; A4, 1.31 M) were the two least accurate of Experiment 1 (91.27% and 93.45%); the best, SE, has 132 K, and the rank correlation between

parameters and accuracy was -0.55 ($p = 0.13$).

24. **What is the difference between an internal attention map and Grad-CAM?** An attention map is a tensor the module computes in the forward pass and multiplies into the features: it is a part of the model and affects the prediction. Grad-CAM is computed afterwards from gradients of the class score with respect to a feature map; it explains the decision but changes nothing.
25. **Why is Grad-CAM called a post-hoc explanation technique?** Because it is applied to a trained, fixed network after the prediction has been made, using the gradient of the predicted class score, rather than being built into the network.
26. **Does the attention model suppress irrelevant background regions?** Not visibly. In the five-image comparison the A3 Grad-CAM was a little more compact than the baseline's on the tulip field and almost identical on the rose and sunflowers, and the spatial gates were ≈ 1 across the background (Plots 14, 17).
27. **Are the attention maps of correctly and incorrectly classified images different?** The spatial and channel gates are essentially the same for both (they depend little on the image), but the Grad-CAM maps differ: for every model the maps of wrongly classified images are much more compact (coverage 0.10–0.21 versus 0.22–0.33) and slightly less centred (Section 27.1).
28. **Which flower classes benefit most from attention?** For A3 versus the baseline, tulips (recall 95.8% $\rightarrow$ 97.5%, F1 +2.0 pp) and, slightly, roses (F1 +1.0 pp); daisy, sunflowers and dandelion lost 1.0–1.7 pp F1 (Section 25). These are changes of a few images per class.
29. **What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?** Which classes are confused with which, in which direction, and how the errors are redistributed between models: A0 and A3 differ by one image overall, yet A3 gains 2 tulip images and loses 3 daisy images to dandelion.
30. **Why is macro F1 useful in addition to accuracy?** It weights every class equally and combines precision and recall, so it exposes a model that is good on the large classes (dandelion 134 images) and poor on the small ones (daisy 95). Here it moved with accuracy (within 0.3 pp), which itself shows that no model was class-biased.
31. **What is meant by the accuracy–complexity trade-off?** Whether the extra parameters or computation of a module are repaid by a gain in accuracy. In Plot 7 SE gains 0.73 pp for +0.56% parameters, while the 5.6% larger self-attention model loses 1.09 pp: the trade-off is favourable only for the cheap channel module.
32. **What is meant by the F1–inference-time trade-off?** The F1 gained per millisecond of extra inference time. Plot 8: SE costs +0.03 ms (+2.8%) for +0.58 pp of F1; self-attention costs +0.09 ms (+7%) for -1.17 pp; the combined model +0.15 ms (+13%) for -3.4 pp.
33. **Why must every ablation study use the same dataset split?** Different splits change both the training data and the test images, so differences between models would partly reflect the split. With a 550-image test set one image is 0.18 pp, so a different test set could by itself change the accuracy by more than most of the differences in this report.
34. **Why must the same training protocol be maintained when comparing different attention mechanisms?** Learning rate, epochs, augmentation, seed and fine-tuning all influence accuracy at least as much as the module itself; if they changed together with the module, the effect of the attention could not be separated from them.
35. **What conclusions can be drawn from the Grad-CAM comparison?** (i) Baseline and attention models look at the same region of the flower; (ii) mechanisms change the width of the activation (A3, CBAM, spatial: more compact; self-attention, combined, Coordinate: more spread) but not its position; (iii) wrong predictions have more compact maps; (iv) only 1 of the 5 images (the white-and-pink bouquet) changed prediction with the mechanism (SE, CBAM, Coordinate correct; the others wrong), and 1 (rose fence) was wrong for all models. With five images these are observations, not general conclusions.

38. Observations to be Recorded

The values observed for every model are the *restored* (best-validation) epoch's metrics.

Model	Train acc	Val acc	Test acc	Train loss	Val loss	Test loss	Restored epoch
A0: Baseline	0.9988	0.9437	0.9455	0.0049	0.2446	0.2087	14
A1: Channel (SE)	0.9953	0.9492	0.9527	0.0108	0.2494	0.1628	12
A2: Spatial (7×7)	0.9969	0.9492	0.9436	0.0111	0.2239	0.1820	10
A3: Channel→Spatial	0.9984	0.9492	0.9436	0.0061	0.2974	0.2201	12
A4: Self-Attention	0.9965	0.9474	0.9345	0.0103	0.2135	0.2126	13
A5: C→S→Self	0.9560	0.9437	0.9127	0.1239	0.1785	0.2216	5
B1: C→S at P_1	0.9899	0.9256	0.9291	0.0257	0.3146	0.2875	11
B2: C→S at P_2	0.9973	0.9292	0.9473	0.0077	0.3016	0.2501	12
B4: C→S at P_4	0.9977	0.9456	0.9455	0.0054	0.2826	0.2444	13
M34: C→S at P_3+P_4	0.9996	0.9456	0.9436	0.0028	0.2691	0.2592	13
M234: C→S at P_2+P_3+P_4	0.9957	0.9274	0.9364	0.0137	0.3245	0.2449	12
O1: Spatial→Channel	0.9984	0.9419	0.9491	0.0056	0.3031	0.2126	12
K3: C→S, $k = 3$	0.9813	0.9347	0.9327	0.0482	0.3554	0.3035	13
K5: C→S, $k = 5$	0.9969	0.9401	0.9418	0.0143	0.2542	0.2382	11
X1: ECA	0.9973	0.9437	0.9382	0.0088	0.2406	0.2407	12
X2: CBAM	0.9934	0.9401	0.9345	0.0198	0.1978	0.2406	11
X3: Coordinate Attention	0.9840	0.9401	0.9327	0.0401	0.3116	0.2158	12

"Train acc" and "Train loss" are the running averages over augmented training batches of the restored epoch; validation and test values are computed with the final weights of that epoch. Where each recorded observation is reported:

No.	Observation	Reported in
1	input and output feature dimensions of each attention module	Section 13 (tables)
2–4	training, validation, test accuracy	table above, Sections 14–19, Plot 2
5–6	training and validation loss	table above, Plot 3, Plot 4
7–9	macro precision, recall, F1	Sections 14, 19.1, 23
10–11	trainable and total parameters	Sections 13, 23, 33.2
12	average inference time	Sections 14, 22, 33.2
13	confusion matrix	Section 25 (Plot 11)
14	channel-attention weights	Section 28.1 (Plots 12, 13)
15	spatial-attention maps	Section 28.2 (Plots 14a, 14b)
16	self-attention responses	Section 28.3 (Plot 15)
17	Grad-CAM maps	Sections 27, 30, 32 (Plots 17–19)
18	attention maps for different insertion positions	Section 29 (Plot 16)

39. Laboratory Report Structure

The report follows the structure suggested in the manual; the mapping to the sections of this document is:

Suggested part	Sections
Aim	1, 2
Dataset description	3
ResNet50 architecture	4–7
Attention mechanisms	9–12
Proposed experimental architecture	8, 13
Experimental setup	20–22, 34
Attention-type ablation	14
Attention-position ablation	15, 16
Performance results	17–19, 23, 33.1
Complexity analysis	33.2
Training and validation graphs	24
Confusion matrix analysis	25
Grad-CAM analysis	27, 27.1
Internal attention-map analysis	28, 29
Qualitative comparison	30, 32
Additional attention-mechanism comparison	19, 36
Discussion	40
Conclusion	41

40. Discussion

- **Why did a mechanism change performance?** The only clearly distinguishable change is the loss of the combined model A5 (-3.27 pp, $p = 0.006$). The training record explains it: A5's best checkpoint is the end of the frozen phase (epoch 5) and its training accuracy there is only 95.6%, so three untrained modules with gates initialised near 0.5 left the model under-fitted, and fine-tuning conv5_x never improved its validation accuracy. For all other models the changes (-1.6 to $+0.7$ pp) are as large as the run-to-run noise.
- **How does attention location influence feature refinement?** Attention at the shallow, high-resolution P_1 was the worst (-1.64 pp, $p = 0.093$), and it converged slowest (7 epochs to 90% validation accuracy): re-scaling 256 low-level channels before everything else affects all later stages while the backbone is frozen. From P_2 on, the position made no measurable difference.
- **How does feature resolution change attention behaviour?** The spatial gates are ≈ 1 everywhere; the low-value border strip is a few cells wide at every stage, so its extent shrinks in pixels as the grid shrinks and the gate contrast falls from 0.88 (P_2) to 0.37 (P_3) and 0.15 (P_4). At P_4 the gate has only 49 cells and the maps differ most between images (correlation 0.30).
- **How does model complexity change with different mechanisms?** Parameters: 5 (ECA) to 1.44 M (A5), time: +2.8% to +12.7%, training time unchanged (14–14.5 min). The self-attention models cost the most in all three; the channel/spatial modules are cheap (Section 33.2).
- **Do the improvements justify the cost?** There is only one possible improvement (SE, +0.73 pp, not significant). It is cheap, so it is the only mechanism that gives a favourable accuracy–cost trade-off, but the evidence is weak. The expensive mechanisms (self-attention, combined) did not improve anything.
- **Do attention maps correspond to meaningful object regions?** No. The spatial gates are almost constant with a border artefact (Section 28.2 and 29), the channel gate is almost the same for every image (Section 28.1) and self-attention collapses onto one location (Section 28.3).
- **Does attention help suppress the background?** Not visibly: the Grad-CAM maps of the baseline and of the attention models have similar spread and the same centre bias (0.45–0.48 centre-box share), and the spatial gates are ≈ 1 on the background (Plots 14, 17).
- **Does attention behave differently for incorrect predictions?** Grad-CAM maps are much more compact for wrongly classified images for every model (coverage 0.10–0.21 versus 0.22–0.33); the internal gates are almost image-independent and look the same for both. One of the two incorrect images changed its prediction with the mechanism, one did not.
- **Do quantitative and qualitative observations support each other?** Yes, in a negative sense: the accuracy results show no reliable benefit and the maps show nothing resembling object-focused attention. Higher compactness of a Grad-CAM did not mean higher accuracy.

Limitations. The conclusions should be read with the following in mind.

1. **One run per model.** Each configuration was trained once. GPU operations were not made deterministic, so a rerun could flip a few test images; differences of a few images (0.18 pp each) between models cannot be ranked.

A fair answer to "which is best" would need several seeds.

2. **Test-set resolution.** With 550 images the 95% interval of any accuracy is about ±2 pp (Section 33.1).
3. **Initialisation of the gates.** Sigmoid gates were initialised randomly (outputs near 0.5), not near identity as is sometimes done; this probably penalises shallow and multi-stage placements and the combined module and favours self-attention (initialised at identity). A different initialisation could change the ranking and was not tried.
4. **Short fine-tuning.** Only conv5_x was fine-tuned, for 10 epochs; modules at P_1–P_3 sit between frozen backbone stages. The conclusions concern this transfer-learning protocol only.
5. **Coarse Grad-CAM.** The maps have 7×7 resolution (each cell 32×32 pixels); statements about petals or the flower centre are limited by this.
6. **Selection of images.** The qualitative comparison uses five images (three correct, two incorrect); the quantitative Grad-CAM statistics over all 550 test images are therefore reported alongside.

41. Expected Outcome and Conclusion

At the end of the experiment, an attention mechanism should be understood not as an isolated module whose behaviour is independent of the backbone. Its effectiveness depends on

$$\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity.}\end{aligned}$$

Classification performance alone does not give a complete explanation of an attention mechanism, so a comprehensive analysis combines

$$\boxed{\text{Classification Performance}} \quad \text{with} \quad \boxed{\text{Computational Complexity}} \quad \text{and} \quad \boxed{\text{Qualitative Visualization.}}$$

Therefore

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions are based on the experimental results rather than on assumptions about which attention mechanism should perform best.

What was measured. Seventeen ResNet50 models were fine-tuned on tf_flowers under one protocol and evaluated on 550 test images.

- **Type (Experiment 1).** Channel attention (SE) is the best model (95.27% accuracy, 95.17% macro F1, +0.73 pp, $p = 0.45$) at +0.56% parameters; spatial and C→S attention equal the baseline (94.36% vs. 94.55%); self-attention (93.45%) and the combined module (91.27%, $p = 0.006$) are worse while being the most expensive.
- **Position (Experiment 2).** Attention at P_1 is worst (92.91%); P_2, P_3, P_4 (94.73%, 94.36%, 94.55%) match the baseline.
- **Multiple positions (Experiment 3).** More positions did not help (P_3+P_4 94.36%, P_2+P_3+P_4 93.64%).
- **Order (Experiment 4).** S→C 94.91% versus C→S 94.36%: 3 images, not significant.
- **Kernel size (Experiment 5).** $k = 3$, 5, 7: 93.27%, 94.18%, 94.36%.
- **Additional mechanisms.** ECA 93.82% (5 parameters), CBAM 93.45%, Coordinate Attention 93.27%: none improves on the baseline.
- **Complexity.** Modules cost +2.8% to +12.7% inference time and 5 to 1.44 M parameters; larger modules were not more accurate ($\rho = -0.55$).
- **Visualisation.** The learned gates are nearly constant (channel gate correlation 0.98–0.998 across images; spatial gates ≈ 1 with a border artefact; self-attention on one hot spot per image); Grad-CAM maps of baseline and attention models look at the same region of the flower.

Conclusion. On this dataset, with a strong pretrained ResNet50 and a short fine-tuning protocol, adding attention did not give a gain that can be separated from run-to-run noise. The only mechanism that may help is the cheap SE channel gate (+4 test images); expensive mechanisms, attention at the earliest stage and stacking several modules do not pay off or lower accuracy. The visualisations explain why: the learned attention is close to constant and not object-selective. The accuracy–complexity trade-off therefore favours light-weight or no attention for this task.

42. Source Code

The complete code for this experiment is available on GitHub:

https://github.com/K-Mithran/Deep-Learning/Lab8

The folder contains the executed notebook `Experiment_8_9.ipynb` (with all outputs), which covers Sections 3–38 of this report: data loading and the stratified split, the augmentation pipeline, the attention modules (channel, spatial, self-attention, ECA, CBAM, Coordinate Attention), the stage-by-stage ResNet50 builder, the two-phase training of all 17 models, the evaluation (metrics, confidence intervals, McNemar tests, timing), and all figures of Plots 1–19 (Grad-CAM, channel, spatial and self-attention maps). The folder also contains this report (`Experiment_8_9_Report.tex` and `.pdf`) and its figures in `figs/`.

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